Trump, Trade, and Emissions: Green Protection vs. Brown Sourcing *

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Can protectionism by a clean economy be climate policy? We study the U.S. tariffs of 2025, the clearest instance of geoeconomic fragmentation so far, in a multi-country, multi-sector trade model with global input-output linkages. Global emissions fall in every scenario, but mainly because the world becomes poorer. The one margin on which a tariff could be the right climate instrument is sourcing, the country a good comes from. We show analytically that a tariff on dirty origins makes sourcing cleaner in the protected market (green protection) but dirtier in all other markets, since higher input costs erode the competitiveness of the clean country abroad (brown sourcing). Both forces appear in every scenario. Sourcing lowers emissions under the tariffs as applied, but this gain rests on the U.S. market alone. When partners retaliate across the board, the U.S. market shrinks and the net sourcing effect turns brown.

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1 Introduction

Can protectionism be climate policy? The world economy is fragmenting along geopolitical lines, and the tariff has returned as an instrument of choice. A common argument states that protectionism can help the climate when the protecting economy is clean, because production returns from dirty producers abroad to cleaner ones at home. During the 2024 U.S. vice presidential debate, J.D. Vance argued that reshoring American manufacturing would help fight climate change on the grounds that the United States runs the cleanest economy.[1] The U.S. tariffs of 2025 put this argument to a test. They are large, they are documented line by line, and they are imposed by one of the cleanest large producers in the world. They are also likely to last. When the Biden administration took office in 2021, it kept most of the tariffs of the first Trump administration, including those on China, and raised some of them in 2024, for example on electric vehicles (Guardian, 2024). Our January 2025 baseline therefore already contains these earlier measures.

The argument is a claim about where goods are produced, not about how much is produced. A tariff can lower emissions because it makes the world poorer (scale) and because it shifts spending toward cleaner sectors (composition). For both targets a domestic instrument such as a carbon tax does better, since the tariff reaches them only with an additional distortion attached (Bhagwati and Ramaswami, 1963). The one thing only trade policy can do is discriminate by origin. The question whether tariffs can be climate policy is therefore a question about sourcing: which country supplies a given good. We argue that the answer depends on the market. A tariff on inputs from a dirty origin raises the production costs of the clean country. In its home market the tariff wedge dominates this cost increase, and the dirty origin loses market share (green protection). In every other market there is no wedge, only the higher cost, so the dirty origin gains (brown sourcing). Which force dominates globally is a quantitative question. Our theory does not depend on which country is the clean one, but our quantitative results speak only to U.S. tariffs.

This paper asks whether broad-based tariffs imposed by a relatively clean economy make global production cleaner or dirtier. We evaluate the U.S. tariffs in a multi-country, multi-sector general equilibrium trade model with global input-output linkages in the tradition of Caliendo and Parro (2015). The model is calibrated to 2018 OECD ICIO data covering 64 countries and regions and 42 industries, and we measure all effects relative to the tariffs in force in January 2025. Emissions are attributed to each country-sector pair at its observed intensity. We model thirteen schedules: four campaign proposals and their retaliatory equivalents, and the implemented Liberation Day, Tariff Peak[2] and Tariff Deal[3] schedules, the last two also

[1] Full transcript available at https://www.cbsnews.com/news/full-vp-debate-transcript-walz-vance-2024/.

[2] Tariff Peak represents the height of the trade war, with average tariff increases on goods of 100% on China and 10% on all other partners.

[3] Tariff Deal represents a moderate scheme after some rounds of negotiations, with average tariff increases on goods of 20% on China and 12% on all other partners.

with their retaliation cases.

To separate the channels through which tariffs move emissions, we use a decomposition in the spirit of the scale-composition-technique literature (Grossman and Krueger, 1991; Copeland and Taylor, 2003) and closest in form to Le Moigne et al. (2024). Emissions change because the world produces less (scale), because it produces a different mix of goods (composition), or because a given good is produced in a different country (sourcing). The aggregate sourcing term is silent on where the reallocation happens. We therefore split the sourcing term exactly into within-market components for the U.S., the E.U., and all other destinations, plus a destination-mix component that captures the changing relative size of markets.

We also derive the mechanism analytically. In a stripped-down two-country version of the model, a tariff imposed by the clean country passes into its own producer price with a pass-through that depends only on the materials share and the share of dirty inputs in its intermediate bundle. Proposition 1 shows that the dirty origin's share falls in the tariff-imposing market (green protection) and rises in the third market (brown sourcing). It also gives the exact condition under which the second force dominates globally. Evaluated at U.S. input shares, unilateral tariffs sit just below the threshold. The theory therefore predicts that brown sourcing in third markets occurs but is smaller than green protection in the U.S. market. Corollary 1 shows that retaliation strengthens brown sourcing, because a retaliatory tariff adds a direct wedge against the clean exporter abroad on top of the cost channel.

Our answer is that the optimistic argument holds in the net for the tariffs as applied, but for a fragile reason. Global emissions fall in every scenario, by between 120.5 and 289.8 Mt CO_2eq. This is comparable on average to the annual emissions of the Netherlands and worth on the order of \$22.9 to \$55.1 billion per year in avoided damages at the EPA's central social cost of carbon. The decomposition shows that tariffs lower emissions mainly by making the world poorer. The scale effect accounts for 62 to 80% of the reduction and the composition effect for 17 to 35%. This is the opposite of how efficient climate policy works. Le Moigne et al. (2024) find that the sourcing margin delivers roughly one third of the emissions reduction achieved by carbon taxes. Under the tariffs, the sourcing effect lowers emissions in 9 of the thirteen scenarios, including all five implemented schedules, and raises them only in the four retaliatory campaign scenarios.

This net sourcing effect hides two large opposing flows. Under Liberation Day, sourcing within the U.S. market lowers emissions by 44.5 Mt. Sourcing within the E.U. market and in all other destinations raises them by 1.2 and 7.2 Mt, and the destination mix raises them by a further 2.3 Mt. This leaves a net of -33.9 Mt. The green side of the ledger therefore rests on a single market. In the retaliatory campaign schedules the within-market terms still sum to a negative number. The net sourcing term turns positive, at up to $+20.5$ Mt, because retaliation shrinks the U.S. share of world demand, and the U.S. market sources more cleanly than the world average.

We test Proposition 1 by observing the within-sector market share of carbon-intensive origins, both in the U.S. market and in third markets. The share of carbon-intensive origins in the U.S. market falls in every one of the thirteen scenarios, by between 0.12 and 0.70 percentage points. The third-market share rises in every scenario, by 0.11 to 0.24 percentage points under unilateral action and by up to 0.29 percentage points where partners retaliate. This confirms green protection, brown sourcing, and the strengthening of brown sourcing under retaliation. We then ask whether brown sourcing runs through input costs. Splitting the within-market sourcing term, we find that the direct tariff wedge lowers emissions in all 13 scenarios, while the induced rise in producer prices raises them in all 13. Under two alternative sets of trade elasticities, the aggregate sourcing term, the market-level terms, and the dirty-origin shares keep their signs.

Territorial and consumption-based accounting tell different halves of the story. Under Liberation Day, U.S. territorial emissions fall by around 11.7 Mt, while the emissions embodied in U.S. final demand fall by around 199.8 Mt. This fall comes from sourcing rather than from lower U.S. consumption. A territorial accounting therefore understates the global reach of the tariffs by a factor of around 17.1. More tellingly, several third countries record rising footprints even as their territorial emissions fall, including India, Canada, Indonesia, and Australia. Their own consumption has become more carbon-intensive, which is the consumption-side signature of brown sourcing.

The emissions results are driven entirely by tariff-induced trade reallocation, so they are only as credible as those flows. We therefore show that the model reproduces the economic effects these policies are known to have. Under the campaign schedules, the United States gains in every unilateral case but one (US1060EU0[4]), by around 0.27% of real income under a uniform 10% tariff. It loses in the retaliation cases (around zero under US10Retal[5]), which is the classic optimal-tariff logic. Among the implemented schedules, the United States gains under Liberation Day (+0.09%) and Tariff Deal (+0.16%) but loses under Tariff Peak (−0.10%), even without retaliation. The difference is dispersion, not severity. Tariff Peak raises the average tariff on goods by a similar amount as Liberation Day (23% against 20%). But it pairs 100% on China with 10% on all other partners. Displaced demand is diverted to third countries rather than reshored, and the United States captures a wedge of 10% instead of 100%. The U.S. gains rest entirely on tariff revenue. Once it is not returned to households, every gain reverses.

This paper makes three contributions. First, we provide a quantitative assessment of the environmental effects of the U.S. tariffs of 2025 as they were applied. We build each schedule line by line and evaluate it against the trade-war baseline of January 2025. Second, we identify the mechanism through which tariffs by

[4]US1060EU0: The U.S. implements 10% on all countries, 60% on China, the EU is exempt, and no country retaliates.

[5]US10Retal: The U.S. implements 10% on all countries, and each country retaliates symmetrically.

a clean economy move global sourcing. Third, we split the sourcing effect by destination market.

These contributions relate to four strands of the literature. The first studies trade policy as climate policy. Shapiro (2021) documents that existing tariff schedules are systematically lower on polluting industries and therefore implicitly subsidize carbon, and Farrokhi and Lashkaripour (2025) and Farrokhi et al. (2025) show how trade instruments could be designed to support climate objectives. That work asks how trade policy might be reformed to help. We ask what happens to the climate when trade policy moves in the direction it is actually moving.

The second is the literature on the recent trade wars, which has established near-complete pass-through to U.S. consumers (Amiti et al., 2019, 2020; Cavallo et al., 2021; Flaaen et al., 2020; Houde and Wang, 2023) and substantial welfare losses (Fajgelbaum et al., 2020b; Fajgelbaum and Khandelwal, 2022; Clausing and Lovely, 2024; Ignatenko et al., 2025). That literature is almost silent on emissions. We add the environmental dimension and show that dispersion across partners can flip the sign of the U.S. welfare effect.

The third is the literature on the environmental gains from trade (Shapiro, 2016; Le Moigne et al., 2024). Those papers measure what open markets deliver. We measure what closing them does to emissions, and we show that the net change in global emissions understates how far tariffs rearrange the carbon content of trade.

The fourth is the literature on carbon leakage. Unilateral carbon pricing raises the costs of regulated producers and shifts market share toward unregulated and often dirtier competitors (Fowlie, 2009; Böhringer et al., 2012). Tariffs on imported inputs work through a similar competitiveness channel, since they raise the costs of downstream producers who also export (Handley et al., 2025). We show that trade policy can generate leakage of its own, without any climate policy in place.

The remainder of the paper is organized as follows. Section 2 details the scenarios analyzed in the paper. Section 3 sets out the model, the decomposition, and Proposition 1. Section 4 covers calibration and data. Section 5 establishes the economic effects and validates the model against the existing literature. Section 6 presents the environmental results and tests the mechanism. Section 7 reports robustness to alternative elasticities, and Section 8 concludes.

2 Choice of scenarios

We analyze three sets of scenarios, summarized in Table 1. The first two sets are based on statements Trump made during his 2024 campaign. In the first set, the U.S. imposes tariffs unilaterally. The second set adds a retaliation equivalent to each of these scenarios, in which U.S. trading partners respond with tariffs on the U.S. The third set covers tariff schedules that were implemented in 2025. All scenarios start from the same

baseline, the tariffs in force in January 2025. Every scenario tariff is added on top of this baseline. Tariffs between third countries stay at their baseline levels throughout.

The first campaign scenario, US10, assumes that the U.S. imposes a 10% tariff on all countries and all sectors (Business, 2024). In its retaliation equivalent, all other countries respond with a 10% tariff on the U.S. The second scenario, US1060, adds a 60% tariff on China, another policy Trump frequently mentioned during the campaign (CNBC, 2024). In the retaliation case, all countries impose a 10% tariff on the U.S., and China retaliates with 60%. The third scenario, US1060EU0, exempts the EU from the trade war. All other countries face a 10% tariff and China faces 60%. Trump has not proposed this exemption explicitly. It allows us to study the effects on countries outside the EU, such as Canada or Mexico. The fourth scenario, US2060, raises the base tariff to 20% and keeps the tariff on China at 60% (Today, 2024). In the retaliation cases of the third and fourth scenarios, each country imposes the tariff it faces from the U.S.

The third set contains three schedules from 2025 (last five rows of Table 1). Liberation Day (LibDay) is the schedule announced in the Rose Garden on 2 April 2025 (Executive Office of the President, 2025), as it entered into force on 9 April. It imposed a baseline tariff on nearly all imports together with country-specific rates of up to 49% among the countries in our data. The country-specific rates were in force for one day. On the same day, Trump paused them for 90 days for all countries except China, effective 10 April. Tariff Peak (TariffPeak) is the schedule on 12 May 2025, at the height of the escalation between the U.S. and China. Tariff Deal (TariffDeal) is the schedule on 17 November 2025, after several rounds of negotiations and bilateral deals. The rates applied over 2025 ranged from full exemption for some goods to 145% on China. Each of these scenarios is the full U.S. tariff schedule on that date, which we build line by line from the Harmonized Tariff Schedule and aggregate to our sectors with 2018 trade weights. Table 2 reports the resulting average tariff by partner. Appendix C summarizes the construction, and the Online Appendix documents every measure. Tariff Peak and Tariff Deal each have a retaliation equivalent with the retaliatory tariffs in force on the same date. In practice, only China and Canada retaliated in 2025.

Table 1: Overview of scenarios

Name	**Description**
US10	US levies a 10% tariff on all countries and all sectors
US1060	US levies a 10% tariff on all countries and all sectors, and a 60% tariff on China
US1060EU0	as above, and the EU can negotiate an exemption (0% tariff)
US2060	US levies a 20% tariff on all countries and all sectors, and a 60% tariff on China
US10Retal	US levies a 10% tariff, each country retaliates with 10% against US
US1060Retal	as above, just US-China both implement 60% against each other
US1060EU0Retal	as above, but exemption for EU (0 tariff between US and EU)
US2060Retal	US levies a 20% tariff, each country retaliates with 20% against US
LibDay	Liberation Day schedule as implemented (9 April 2025)
TariffPeak	Peak of trade war (12 May 2025), only US imposes tariffs
TariffDeal	After several months of negotiation (17 November 2025), only US imposes tariffs
TariffPeakRetal	Peak of trade war, with retaliation
TariffDealRetal	After several months of negotiation, with retaliation

Notes: The first eight scenarios are hypothetical schedules based on statements made during the 2024 campaign. They apply a uniform ad valorem tariff across all sectors. The last five correspond to schedules that were implemented in 2025. Table 2 reports their average tariffs by partner. The suffix "Retal" denotes the retaliation equivalent of a scenario, in which U.S. trading partners impose tariffs on the U.S. In the campaign scenarios, retaliation is symmetric. In the 2025 scenarios, it reproduces the retaliatory tariffs in force on the scenario date. All scenario tariffs are added on top of the January 2025 baseline. Tariffs between third countries stay at their baseline levels throughout.

Table 2: Tariff increases on US goods imports by partner (%)

Partner	Imports ($bn)	Jan 2025	LibDay	TariffPeak	TariffDeal
China	298	14	52	100	20
European Union	364	0	15	9	13
Mexico	288	0	14	14	10
Canada	251	0	10	10	9
Japan	117	0	23	16	16
Korea	72	0	22	12	14
Vietnam	43	0	43	10	19
Cambodia	2	0	48	10	19
India	63	1	20	8	37
United Kingdom	40	0	11	11	8
Switzerland	38	0	14	5	7
Brazil	28	0	9	9	26
Russia	14	16	0	1	1
All partners	1992	3	20	23	13
All partners except China	1694	0	15	10	12

Notes: LibDay, TariffPeak, and TariffDeal report the average increase in the U.S. tariff relative to January 2025, which is the shock the model receives. Jan 2025 reports the average tariff in force on 1 January 2025 on top of the 2018 WITS tariffs. All columns are averages over goods sectors. Each tariff column is weighted by U.S. imports before its own shock. The three scenario columns use the January 2025 baseline, which the Imports column reports. The Jan 2025 column uses 2018 imports in the model. Averages differ from headline rates because of exempt goods, stacked measures, and USMCA preferences. The Online Appendix reports these tariffs relative to 2018 and weighted by 2018 Census trade values. On those weights the increases for all partners are higher, since China has a larger weight in 2018.

3 Model

3.1 Setup

We use a quantitative trade model with many countries, many sectors, and input-output linkages (Armington, 1969; Caliendo and Parro, 2015).

There are N countries indexed by i (for origin) and j (for destination) and S industries indexed by s' (for upstream) and s (for downstream). Each country produces a unique variety within each industry. Trade is subject to iceberg trade costs $\tau_{is'j} \geq 1$, with $\tau_{is'i} = 1$ for all i, and to ad valorem import tariffs $t_{is'j}$. Each country is endowed with an inelastic supply of workers L_i, who cannot move across countries.

3.2 Equilibrium

Consumption choices are made by representative households with Cobb-Douglas-CES preferences

$$U_j = \prod_{s'} \left(U_{s'j}\right)^{\beta_{s'j}} \tag{1}$$

$$U_{s'j} = \left[\sum_i \left(a_{is'}\right)^{1/\sigma_{s'}} \left(q_{is'j}\right)^{(\sigma_{s'}-1)/\sigma_{s'}}\right]^{\sigma_{s'}/(\sigma_{s'}-1)}, \tag{2}$$

where $\beta_{s'j}$ are expenditure shares, $a_{is'}$ are demand shifters, $\sigma_{s'}$ are substitution elasticities, and $q_{is'j}$ are the final consumption quantities of varieties differentiated by country of origin. As a result, household final demand is given by

$$q_{is'j} = a_{is'} \frac{\left[p_{is'j}\left(1+t_{is'j}\right)\right]^{-\sigma_{s'}}}{\left(P^c_{s'j}\right)^{1-\sigma_{s'}}} \beta_{s'j} I_j \tag{3}$$

$$I_j = w_j L_j + R_j + D_j, \tag{4}$$

where $p_{is'j}$ are delivered prices, $t_{is'j}$ are tariffs, $P^c_{s'j}$ are consumer price indices, $w_j L_j$ is labor income, R_j is tax revenue, and D_j is an exogenous transfer used to match aggregate trade deficits in the data, which we keep constant in our counterfactuals. Tariff revenue R_j is rebated to households as a lump sum.

Firms produce these varieties under perfect competition from labor and intermediate goods using Cobb-Douglas-CES technologies

$$q_{js} = A_{js} \left(\frac{L_{js}}{\gamma_{j,L_s}}\right)^{\gamma_{j,L_s}} \prod_{s'} \left(\frac{m_{s'js}}{\gamma_{s'js}}\right)^{\gamma_{s'js}} \tag{5}$$

$$m_{s'js} = \left[\sum_i \left(b_{is'js}\right)^{1/\eta_{s'}} \left(m_{is'js}\right)^{(\eta_{s'}-1)/\eta_{s'}}\right]^{\eta_{s'}/(\eta_{s'}-1)}, \tag{6}$$

where A_{js} are total factor productivities, $\gamma_{s'js}$ are cost shares, $b_{is'js}$ are demand shifters (which can differ across buying countries and sectors, so that the model matches the observed sourcing of each sector exactly), $\eta_{s'}$ are substitution elasticities, and $m_{s'js}$ are the intermediate consumption quantities of the same varieties also demanded by households. As a result, firm intermediate demand is given by

$$m_{is'js} = b_{is'js} \frac{\left[p_{is'j}\left(1+t_{is'j}\right)\right]^{-\eta_{s'}}}{\left(P^{p}_{s'j}\right)^{1-\eta_{s'}}} \gamma_{s'js} E_{js}, \tag{7}$$

where $P^{p}_{s'j}$ are producer price indices.

We close the model by imposing labor and goods market clearing:

$$\sum_{s} L_{js} = L_j \tag{8}$$

$$\underbrace{\sum_{s'}\sum_{j} p_{is'j}\left(q_{is'j} + \sum_{s} m_{is'js}\right)}_{\text{exports of } i} + D_i = \underbrace{\sum_{s'}\sum_{j} p_{js'i}\left(q_{js'i} + \sum_{s} m_{js'is}\right)}_{\text{imports of } i}. \tag{9}$$

D_i represents the trade deficit, which is exogenous and held fixed in our counterfactuals.

We calculate emissions by multiplying the counterfactual quantity of output of each country-sector pair, valued at baseline prices, with its emissions intensity per dollar of output:

$$GHG_{is'} = e_{is'} \sum_{j} p^{0}_{is'j}\left(q_{is'j} + \sum_{s} m_{is'js}\right),$$
$$e_{is'} = \frac{CO2_{is'}}{Y_{is'}},$$

where $e_{is'}$ is the exogenous emissions intensity of sector s' in country i, in tons of CO_2eq per dollar of output, $Y_{is'}$ is gross output, and $p^{0}_{is'j}$ are baseline prices. Global emissions are $GHG = \sum_i \sum_{s'} GHG_{is'}$.

The intensity $e_{is'}$ captures only the direct emissions from the production process at each stage of the supply chain. The *total* emissions embodied in a good therefore still respond to tariffs, as firms change where they source their inputs. Farrokhi and Lashkaripour (2025) and Garcia-Lembergman et al. (2025) also make direct emissions endogenous, so that firms can adjust abatement in response to policy. Since tariffs do not target emissions directly, we expect this abatement channel to be of second order. We therefore abstract from it and focus on the main channel through which tariffs affect emissions: the reorganization of global production and sourcing.

3.3 Decomposition

To analyze how trade policy affects greenhouse gas (GHG) emissions, we decompose the change in emissions along the lines of the scale, composition, and technique effects used in environmental economics (Grossman and Helpman, 1995; Copeland and Taylor, 2003; Levinson, 2009; Shapiro and Walker, 2018). We follow Le Moigne et al. (2024), who use a similar decomposition to study the environmental gains from trade under efficient climate policies.

We write total emissions as $GHG = \sum_i \sum_j \sum_{s'} q_{is'j}\, e_{is'}$, where $q_{is'j}$ is the quantity of goods flowing from country i in sector s' to country j and $e_{is'}$ is the emissions intensity per quantity. Write $q_{is} = \sum_j q_{isj}$ for country–sector output, and let $E_{is} \equiv e_{is} q_{is}$, $E_s \equiv \sum_i E_{is}$, and $E \equiv \sum_s E_s = GHG$ denote emissions levels at the country–sector, sector, and global level.

Totally differentiating this expression holding emissions intensities constant, we decompose the overall change in emissions into a scale, composition, and sourcing effect:

$$d \ln GHG = \underbrace{d \ln q}_{\text{scale effect}} + \underbrace{\sum_s \frac{E_s}{E} d \ln \frac{q_s}{q}}_{\text{composition effect}} + \underbrace{\sum_i \sum_s \frac{E_{is}}{E} d \ln \frac{q_{is}}{q_s}}_{\text{sourcing effect}}, \tag{10}$$

where $\frac{E_s}{E}$ is the emissions share of a sector, $\frac{q_s}{q}$ is the trade volume share of sector s, $\frac{E_{is}}{E}$ is the emissions share of a country-sector pair, and $\frac{q_{is}}{q_s}$ is the trade volume share of a country-sector pair in the trade volume of that sector.

Applied to our model, equation (10) captures three channels through which trade policy affects emissions. First, tariffs make all goods more expensive and thus reduce aggregate expenditure. This is the scale effect. Second, tariffs reallocate expenditure across industries. Since some industries are cleaner than others, emissions can rise or fall. This is the composition effect. Third, tariffs reallocate expenditure within a sector across countries of origin, whose emissions intensities differ. This is the sourcing effect. While the scale and composition effects also occur in a closed economy, the sourcing effect is specific to international trade. Our decomposition has no technique effect, since emissions intensities are constant.

The sourcing effect is the net of two much larger opposing forces. It also does not tell us *where* the reallocation happens. The share q_{is}/q_s of a country-sector pair can rise because it gains ground in the U.S. market, in the EU market, or in all other markets. We therefore split the sourcing effect further by destination market.

We partition destination countries j into three regions $r \in \{\text{US}, \text{EU}, \text{Other}\}$, where Other contains every

destination outside the U.S. and the EU. Define

$$q_{isr} = \sum_{j \in r} q_{isj}, \qquad q_{sr} = \sum_{i} q_{isr}, \qquad \mu_{sr} = \frac{q_{sr}}{q_s}, \qquad \tilde{s}_{isr} = \frac{q_{isr}}{q_{sr}}, \tag{11}$$

so that μ_{sr} is the share of region r in world trade of sector s, and $\tilde{s}_{isr}$ is the share of origin i in the sector-s purchases of region r. By construction, q_{is}/q_s is a weighted average of these market-specific shares, with the market sizes as weights:

$$\frac{q_{is}}{q_s} = \sum_{r} \mu_{sr}\, \tilde{s}_{isr}. \tag{12}$$

Applying the product rule to $\Delta(\mu_{sr}\tilde{s}_{isr})$ and summing over r splits the change in q_{is}/q_s exactly into a within-market and a destination-mix component:

$$\Delta\left(\frac{q_{is}}{q_s}\right) = \underbrace{\sum_{r} \mu^{bl}_{sr}\, \Delta\tilde{s}_{isr}}_{\text{within-market}} + \underbrace{\sum_{r} \left(\tilde{s}^{bl}_{isr}\, \Delta\mu_{sr} + \Delta\mu_{sr}\, \Delta\tilde{s}_{isr}\right)}_{\text{destination mix (incl. interaction)}}. \tag{13}$$

To report the sourcing effect in levels, we drop the normalization by E in the last term of equation (10). This gives $GS = \sum_i \sum_s E_{is}\, d\ln(q_{is}/q_s) \approx \sum_i \sum_s \frac{E_{is}}{q_{is}/q_s}\, \Delta(q_{is}/q_s)$. Substituting the split above and summing over i and s gives

$$GS = \sum_{r} GS_r + MIX, \qquad GS_r = \sum_{i} \sum_{s} \frac{E_{is}}{q_{is}/q_s}\, \mu^{bl}_{sr}\, \Delta\tilde{s}_{isr}, \tag{14}$$

where MIX collects the destination-mix and interaction terms in the same way. The only approximation is the step from $d\ln$ to Δ in GS. Given GS, the split into regions is exact.

The term GS_r holds the size of market r at its baseline value. It measures how much the emissions-weighted global share of origin i moves because its share *within* market r changes. For example, a dirty origin can lose share of U.S. purchases because of the tariff, or gain share of EU purchases through trade diversion. The term MIX holds every within-market share fixed. It measures how much the global share of an origin moves only because the relative *size* of the markets it serves changes. This is an effect of the composition of demand, not of competitiveness.

3.4 Green protection versus brown sourcing: a stylized mechanism

The sign of the sourcing term in equation (10) is not determined a priori, since the term is the net of opposing forces. This subsection isolates the mechanism in a simple version of the model that has a closed-form

solution. We derive the condition under which a tariff imposed by a clean country shifts *global* production toward a dirty one. The proposition gives two predictions, which we take to the data in Section 6.

Environment. Consider the model of Section 3 with one sector and two countries, $i \in \{C, D\}$, where C is cleaner than D: $e_C < e_D$, with e_i the emissions intensity of origin i. We drop the sector subscripts throughout. We write $\alpha \in (0,1)$ for the materials cost share $\sum_{s'} \gamma_{s'js}$, ω_{ki} for the intermediate demand shifters $b_{is'js}$, θ_{ij} for the final demand shifters $a_{is'}$, and $\sigma, \eta > 1$ for the two substitution elasticities. Production is $Y_i = A_i L_i^{1-\alpha} M_i^{\alpha}$, where M_i is a CES aggregate of intermediates from both countries. Final demand in each destination j is the CES aggregate X_j over the two origins. Country C imposes a small ad valorem tariff $t > 0$ on shipments from D. As in equations (3) and (7), the tariff applies to final and intermediate shipments alike.

We make three simplifications. First, we hold the final demand quantity indices X_C and X_D fixed. This shuts down the scale effect and isolates the sourcing margin. Second, we set $\omega_{CD} = 0$, so that the unit cost of D does not depend on p_C. This is not essential, and Appendix E relaxes it. Third, we hold wages fixed. The exercise therefore describes how demand shifts across origins at given factor prices, not a closed general equilibrium.[6] Both parts of the proposition therefore hold at fixed factor prices. If w_D adjusts, it falls, since D loses access to the market of C. This strengthens part (b), because the change in the relative price in market D becomes $d\ln p_D/dt - \Lambda < -\Lambda$. Part (a) still holds as long as $|d\ln p_D/dt| < 1 - \Lambda$, that is, as long as the fall in the price of the exporter does not undo the tariff wedge itself.

Cost pass-through. Let $\bar{\omega}_{DC}$ denote the cost share of intermediates from D in the intermediate bundle of C at $t = 0$, and $\bar{\omega}_{CC} = 1 - \bar{\omega}_{DC}$. The tariff raises the input costs of C, which raises its output price. Since the output of C is itself an input into its own production, this feeds back into its input costs. Solving this fixed point gives the pass-through of the tariff into the factory-gate price of C:

$$\left.\frac{\partial \ln p_C}{\partial t}\right|_{t=0} = \Lambda \equiv \frac{\alpha\,\bar{\omega}_{DC}}{1 - \alpha\,\bar{\omega}_{CC}} \in (0,1). \tag{15}$$

[6] Fixing wages is a partial equilibrium device to isolate the substitution margin. It is not a claim about how a closed version of this economy would adjust. Cost minimization under Cobb-Douglas technology gives $w_i L_i = (1-\alpha) p_i Y_i$. With L_i supplied inelastically, factor market clearing therefore pins down *nominal* gross output given w_i. Real output $Y_i = w_i L_i / [(1-\alpha) p_i]$ still varies with the real wage. Holding wages and the quantity indices X_C and X_D fixed describes the shift of demand across origins at given factor prices. This is the object of the proposition. It is the analogue of the sourcing term in equation (10), which also holds the size of the sector fixed and is separate from the scale term. The full model of Section 3 solves for wages, prices, and quantities jointly.

The feedback through domestic inputs matters for the size of Λ. It exceeds the direct effect $\alpha\bar{\omega}_{DC}$ by the multiplier $(1-\alpha\bar{\omega}_{CC})^{-1}$. The elasticity η does not appear. At $t=0$, the envelope theorem implies that only cost shares matter for the first-order price response.

Shares. Let $\tilde{s}_{Dj} \equiv q_{Dj}/(q_{Cj}+q_{Dj})$ be the quantity share of the dirty country within destination j, and let ς_D be its global quantity share. With μ_j the share of destination j in global quantities, $\varsigma_D = \sum_j \mu_j \tilde{s}_{Dj}$ and

$$d\varsigma_D = \underbrace{\sum_j \mu_j \, d\tilde{s}_{Dj}}_{\text{within-market}} + \underbrace{\sum_j \tilde{s}_{Dj} \, d\mu_j}_{\text{destination mix}} . \tag{16}$$

Since X_C and X_D are fixed, the weights μ_j are fixed, and the proposition below describes the within-market component. This is the object that Figure 6 reports separately from the destination mix.

Proposition 1 (Green protection versus brown sourcing) *In the environment above, evaluating derivatives at $t=0$:*

(a) (Green protection.) *The dirty country's share in the tariff-imposing market falls, $\partial\tilde{s}_{DC}/\partial t|_{t=0} < 0$.*

(b) (Brown sourcing.) *The dirty country's share in the third market rises, $\partial\tilde{s}_{DD}/\partial t|_{t=0} > 0$.*

Let $A \equiv \mu_C \, \tilde{s}_{DC}(1-\tilde{s}_{DC})$ and $B \equiv \mu_D \, \tilde{s}_{DD}(1-\tilde{s}_{DD})$. The within-market component of $d\varsigma_D$ is positive, and the sourcing term of equation (10) is therefore positive, if and only if

$$B\,\Lambda \;>\; A\,(1-\Lambda), \qquad \textit{equivalently} \qquad \Lambda \;>\; \frac{A}{A+B}. \tag{17}$$

The link between the global share ς_D and the decomposition is exact. With two origins in a single sector, the sourcing term of equation (10) equals

$$\sum_i \frac{E_i}{E} \, d\ln\varsigma_i \;=\; \frac{q\,(e_D - e_C)}{E} \, d\varsigma_D. \tag{18}$$

Since $e_D > e_C$, the sourcing term has the same sign as $d\varsigma_D$.

Corollary 1 (Retaliation) *Suppose D retaliates, imposing a tariff t_R on shipments from C delivered into D, with $\rho \equiv \partial t_R/\partial t \geq 0$. Then part (a) is unchanged, part (b) is reinforced, and condition (17) becomes $B(\Lambda+\rho) > A(1-\Lambda)$. Under reciprocal retaliation ($\rho = 1$) this holds whenever $B \geq A$, that is, whenever the third market is at least as large as the tariff-imposing market once shares are weighted by their dispersion.*

Interpretation. Equation (15) is the core of the mechanism. A tariff on inputs from the dirty country raises the production costs of the clean country in proportion to its reliance on those inputs. In the tariff-imposing market, the direct tariff wedge outweighs this cost increase, so the dirty origin loses share. This is green protection, part (a). In the third market there is no wedge, only the cost increase. The clean country loses competitiveness there, and the dirty origin gains share. This is brown sourcing, part (b). Both parts hold for any small tariff. Condition (17) only decides which of the two forces dominates in the global share of the dirty origin.

The condition is informative about magnitudes. To illustrate this, we collapse the January 2025 baseline into the United States and the rest of the world, pool all sectors, and call an origin dirty if its emissions intensity is above the mean across countries. The materials share α is then around 0.44, dirty origins supply 2.9% of U.S. intermediates ($\bar{\omega}_{DC}$), and μ_C is around 0.22. The pass-through Λ is around 0.021. The threshold $A/(A+B)$ is around 0.028. It is this low because dirty origins supply 2.6% of the U.S. market and 50.1% of the rest of the world, which makes B much larger than A. Unilateral tariffs by a large clean economy therefore leave the within-market reallocation on the green side, but only by a small margin. Brown sourcing in third markets occurs, but it is smaller than green protection in the tariff-imposing market. Retaliation strengthens brown sourcing, because it adds a direct wedge against the clean exporter in the third market (Corollary 1). In the two-country model, this wedge applies to the whole market share of the clean country abroad, so reciprocal retaliation reverses the sign of the within-market component whenever $B \geq A$. In the full model, the United States is only one of many suppliers in each third market, and the retaliating countries are not all dirty. Retaliation then pushes the within-market component toward zero but need not reverse its sign. Section 6.4 confirms both parts of the proposition in every scenario. It also shows that the within-market component stays negative under retaliation, and that the positive sourcing term in the retaliation scenarios comes from the destination mix, which the proposition holds fixed.

Two features of the full model are absent from this stylized version. Neither changes parts (a) and (b). First, intermediate sourcing responds with elasticity η alongside final demand with elasticity σ. Since both margins face the same relative price, parts (a) and (b) hold with σ replaced by a flow-weighted average of σ and η. Condition (17) is unchanged if this average is the same in both markets. Otherwise, A and B are reweighted by the market-specific averages. Second, if $\omega_{CD} > 0$, the price p_D responds to p_C. This replaces Λ by a smaller but still positive constant. It weakens both forces and leaves the structure of condition (17) unchanged. Appendix E proves the proposition and both extensions.

4 Calibration

4.1 Methodology

To bring the model to the data, we use the exact hat algebra of Dekle et al. (2007). We write the equilibrium conditions in changes relative to a baseline equilibrium. This way, we do not need to estimate the preference shifters ($a_{is'}$, $b_{is'js}$), the productivity parameters (A_{js}), or the iceberg trade costs ($\tau_{is'j}$). The model also matches the observed global production and trade flows in the baseline exactly.

To compute the counterfactual equilibrium under a given tariff schedule, we reduce the model to a system of $N \times S$ equations and solve it numerically with a nested fixed-point algorithm.[7] The calibration requires an $N \times N \times S$ matrix of final goods trade flows, an $N \times N \times S \times S$ matrix of intermediate goods trade flows, an $N \times S$ vector of greenhouse gas emissions, the tariffs in the baseline, and estimates of the elasticities $\eta_{s'}$ and $\sigma_{s'}$.

Our baseline is the world of January 2025, not the world of 2018 in which we observe trade flows. On 1 January 2025, the U.S. applied additional duties on China, on steel and aluminum, on solar panels, and on Russia, and China, Russia, and Turkey applied retaliatory duties on the U.S. The 2018 tariff data do not contain these measures. We therefore proceed in two steps. First, we calibrate the model to the 2018 data and solve for the equilibrium under the tariffs in force on 1 January 2025. Second, we use this equilibrium as the baseline and solve every scenario relative to it. All effects we report are thus changes relative to January 2025. Appendix C describes the January 2025 tariffs.

4.2 Data

We use data on trade flows, tariffs, and greenhouse gas (GHG) emissions from several sources. Trade flows of intermediate and final goods come from the OECD Inter-Country Input-Output (ICIO) tables for 1995 to 2018. In our model, they cover 42 industries and 64 economies, including a rest of the world. Applied import tariffs for the same years come from the World Bank (WITS). We calibrate the model to 2018 and use all years to estimate the elasticities. The tariffs in force in January 2025 and in the 2025 scenarios add the U.S. measures and the foreign retaliation that the WITS data do not contain. We build them from the Harmonized Tariff Schedule at the product level and aggregate them to our sectors with 2018 trade weights (Appendix C).

Emissions, measured in CO_2 equivalents, combine the OECD TECO2 dataset, FAOSTAT, and the EDGAR database of the European Commission. They cover emissions from fuel combustion, agriculture,

[7] Appendices D and F detail the equilibrium conditions in changes and the solution algorithm.

land use, industrial processes, and fugitive sources. Together, these sources account for around 93% of global GHG emissions. We calibrate the emissions intensity $e_{is'}$ of each country-sector pair as its 2018 emissions divided by its 2018 gross output, and hold it fixed in all counterfactuals. The data report output in values, so we measure quantities at baseline prices. A country-sector pair's change in quantity is its change in output value divided by its change in price, $\hat{q}_{is'} = \hat{Y}_{is'}/\hat{p}_{is'}$. We aggregate these changes to $q_{s'}$ and q with baseline output as weights. Emissions then change one for one with $\hat{q}_{is'}$, and the three terms of the decomposition in equation (10) sum to within around 0.3 Mt of the total change in emissions in every scenario. Appendix B gives details on how we combine the sources.

4.3 Estimation

Following Le Moigne et al. (2024), we estimate the substitution elasticities with the approach of Caliendo and Parro (2015), assuming for simplicity that $\sigma_{s'} = \eta_{s's} = \eta_{s'}$. This method uses a fixed-effects model to infer the elasticities from the response of trade flows to import tariffs over all years in our data. When an estimate is negative, not statistically significant, or cannot be identified because of limited tariff data, as in most service sectors, we replace it with the mean. Appendix G shows that our estimates are within the usual range in the literature. Section 7 solves all scenarios with two alternative sets of elasticities based on Fontagné et al. (2022).

5 Economic effects of trade policy

Before moving to the environmental effects of trade policy, we first need to establish the economic effects of our model. As the environmental effects that follow are driven entirely by the tariff-induced reallocation of trade and production, their credibility rests on the model reproducing the right trade-flow adjustments. This section therefore does two things. First, it validates the model against the established economic findings in this literature. We will refer to Ignatenko et al. (2025) who provide a thorough account of the macroeconomic consequences of the Liberation Day tariffs. Second, it extends the analysis to several other scenarios—announced in the campaign and implemented after the Liberation Day announcements. Tariff Peak and Tariff Deal correspond to the tariffs actually implemented in 2025 and are the policy-relevant cases for the emissions analysis in Section 6.

5.1 Real income across countries - Campaign scenarios

Figure 1 presents real income changes for each unilateral (left panel) and retaliatory (right panel) scenario announced in the campaign before the election. The U.S. gains in every unilateral scenario except US1060EU0[8] (-0.02%) (left panel) while most other countries lose, and each unilateral scenario produces a net global loss. In US10[9] (blue), U.S. real income rises by about 0.27%. Adding a 60% tariff on China (orange) shrinks the U.S. gain and deepens China's loss. In that scenario, Cambodia benefits, capturing part of China's market share. Exempting the EU (green) turns most EU countries positive. The largest losers are partners for whom the U.S. is a major export destination: Canada and Mexico, and Ireland, a tax haven hosting many U.S. affiliates, which falls by 2.73% under US2060 (red).[10] The global decline reflects the contraction in activity from tariff-induced increases in cross-country and cross-sector prices.

The right panel of Figure 1 repeats these scenarios with each country retaliating at an equivalent rate: if the U.S. imposes 60% on China, China imposes 60% on the U.S., while other partners hold their existing rates. The U.S. now loses in every scenario, reversing the unilateral gains, while losses shrink for most partners as retaliation recovers some of the burden. Ireland's US2060 loss, for instance, falls from 2.73% to 2.28% once it retaliates at 20%.

That U.S. real income rises unilaterally and falls under retaliation is the classic optimal-tariff logic: a large country manipulates its terms of trade (the price of its exports relative to its imports) by taxing imports, and smaller exporters absorb part of the burden through lower pre-tariff prices (Johnson, 1953; Ossa, 2014).

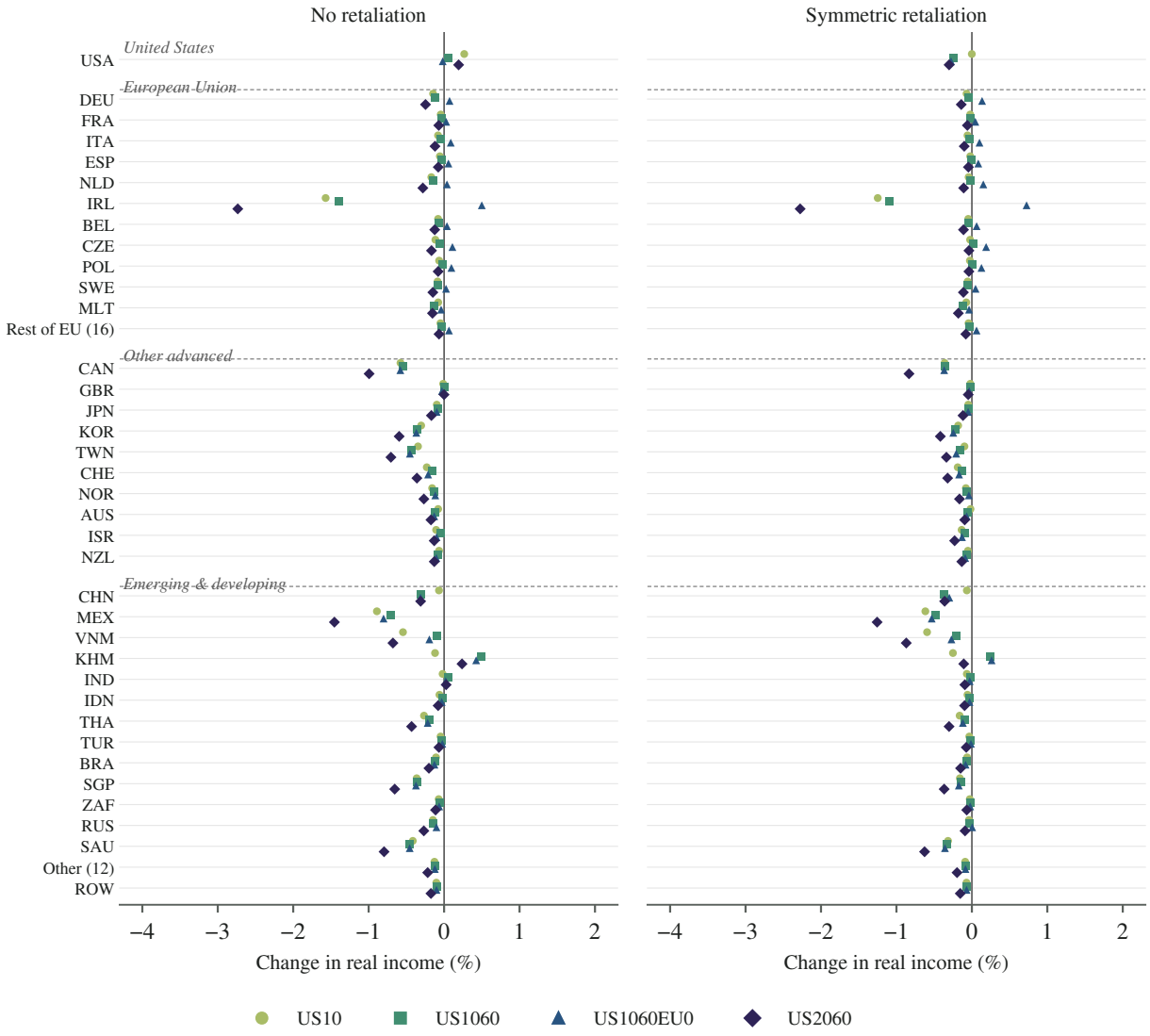
While the tariff schedules in this section differ from the Liberation Day tariffs in Ignatenko et al. (2025), the overall patterns reproduce their central economic findings. We turn to Liberation Day tariffs in the next section. Three features of our setup differ from theirs and are worth stating, since they shape the trade flows that carry the emissions results. We hold the aggregate trade deficit fixed rather than letting it respond to tariffs, so U.S. gains are not eroded by the welfare cost of deficit reduction that operates in their model. Our households supply labor inelastically and we model no income tax, so tariff revenue returns to households only as a lump sum. We thus omit the efficiency gain from using tariff revenue to cut distortionary labor taxes, which accounts for the largest part of the U.S. gain in their baseline. And our competitive, constant-returns structure has no markup or selection margin, so trade responds to a tariff only through substitution across origins, without the additional sensitivity firm selection generates in a Melitz environment. The first

[8] US1060EU0: the U.S. imposes a 10% tariff on all trading partners and a 60% tariff on China, with no retaliation.

[9] US10: the U.S. imposes a 10% tariff on all trading partners, with no retaliation.

[10] US2060: the U.S. imposes a 20% tariff on all trading partners and a 60% tariff on China, with no retaliation.

Figure 1: Real Income Changes Campaign Scenarios



Notes: Change in real income relative to the January 2025 baseline, in percent, under the four campaign scenarios. The left panel reports the unilateral case, in which only the United States raises tariffs; the right panel reports symmetric retaliation, in which each partner imposes on the United States the same rate it faces. Tariffs between third countries are unchanged. Economies are grouped into the United States, EU members, other advanced economies, and emerging and developing economies; the remaining 16 EU members and 12 other economies are shown as single aggregated bars, and ROW is the ICIO rest-of-world residual. Tariff revenue is rebated to households as a lump sum.

two work in opposite directions: the fixed deficit removes a welfare cost their model carries and so raises our gains, while the absence of a distortionary-tax margin removes the channel that accounts for the largest part of the U.S. gain in their baseline and so lowers them. The third, together with a slightly lower trade elasticity, raises our optimal tariff, as shown below.

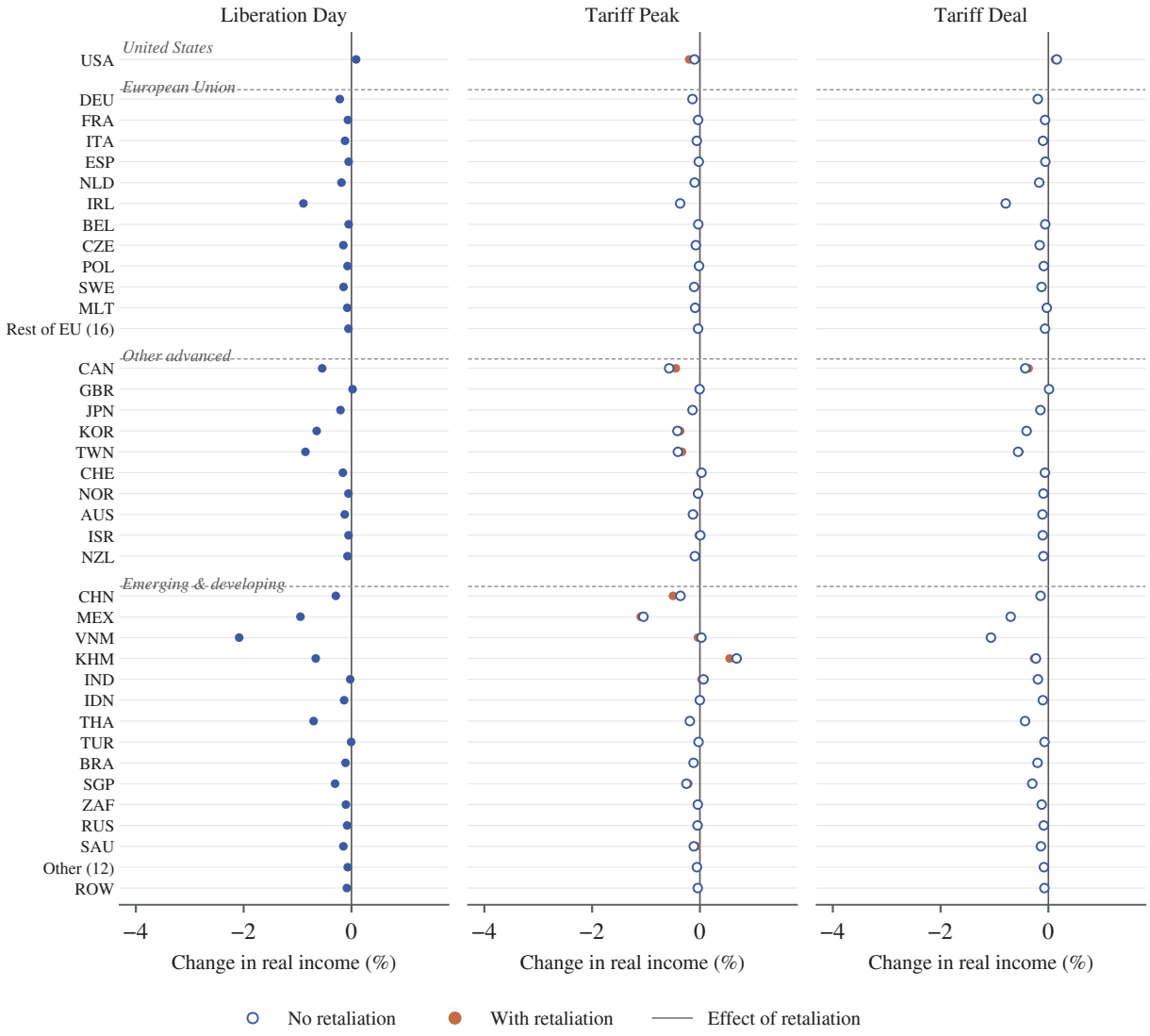
5.2 Real income across countries: implemented scenarios

Figure 2 reports the implemented scenarios (Liberation Day, Tariff Peak, Tariff Deal; Table 2 reports the tariff rates). The U.S. gains under Liberation Day (+0.09%) and Tariff Deal (+0.16%) but *loses* under Tariff Peak (−0.10%), even absent retaliation. The contrast is the section's main economic point. Liberation Day and Tariff Deal raise the tariff on goods by 20% and 13% on average, below the welfare-maximizing uniform tariff of 25% (see Appendix Section F.1). Their rates vary moderately across partners, so the terms-of-trade transfer dominates the distortion. Tariff Peak has an average close to that of Liberation Day (23%), but its rates are far more dispersed. The trade-weighted standard deviation across origins is 24.4 percentage points, against 11.8 under Liberation Day and 4.7 under Tariff Deal. Tariff Peak pairs an average of 100% on China, and up to 145% in some sectors, with 10% on all other partners. At that level the tariff is self-defeating: Chinese import volumes collapse, eroding the tax base and thus the revenue through which the terms-of-trade gain is realized, while the input-cost distortion remains. Because the marginal alternative supplier faces only around 10%, displaced demand is diverted to third countries rather than reshored, and the U.S. captures a wedge of around 10% rather than 100% on the reallocated trade. Dispersion across partners thus converts a potential terms-of-trade gain into a trade-diversion loss.

This sharpens the policy design critique in Ignatenko et al. (2025), who show analytically that the welfare-maximizing tariff is uniform, and that the dispersed United States Trade Representative (USTR) schedule leaves gains on the table. In our implemented schedules the same force is strong enough to flip the sign of the U.S. effect. Because our model has no distortionary income tax, this occurs in the setting that corresponds to their lump-sum rebate case, where no tax-substitution gain is available to offset it.

The relative-price mechanism is visible across Asia: Vietnam and Cambodia lose under Liberation Day (−2.08% and −0.66%, facing 43% and 48%, with Vietnam the largest loser of all), yet gain under Tariff Peak (+0.03% and +0.68%) when both their rates fall to 10% while the rate on China rises to 100%. Mexico and Canada, whose goods exports to the U.S. face average new tariffs of about 14% and 10% in the Liberation Day and Tariff Peak scenarios, and 10% and 9% in the Tariff Deal scenario, lose under all three scenarios. (−0.95% and −0.54% under Liberation Day, −1.05% and −0.57% under Tariff Peak, and −0.70% and −0.43% under Tariff Deal). What matters for third countries is not the level of U.S. protection but their tariff position relative to competing suppliers. This is the same relative-price channel that drives the sourcing reallocation behind our emissions results in Section 6.

Figure 2: Real Income Changes Implemented Scenarios



Notes: Change in real income relative to the January 2025 baseline, in percent, under the implemented schedules (Liberation Day, Tariff Peak, Tariff Deal). Within each panel, the two markers report the scenario without retaliation and with the retaliatory tariffs actually imposed; the connecting segment measures the effect of retaliation. Liberation Day has no retaliation counterpart, so only the unilateral case is shown. Grouping of economies and treatment of tariff revenue are as in Figure 1.

5.3 Tariff revenue in general equilibrium

Because the U.S. real income results run through tariff revenue, we examine how much these policies actually raise. Figure 3 compares a naive calculation, in which the counterfactual tariff schedule is applied to baseline import volumes, with the revenue that arises once quantities adjust. Two patterns emerge. First, realized revenue falls short of the naive benchmark in every scenario, and the shortfall is much larger under the

higher schedules: about 15% under US10 and about 60% under Tariff Peak. This reflects the endogenous contraction of imports the naive calculation holds fixed.

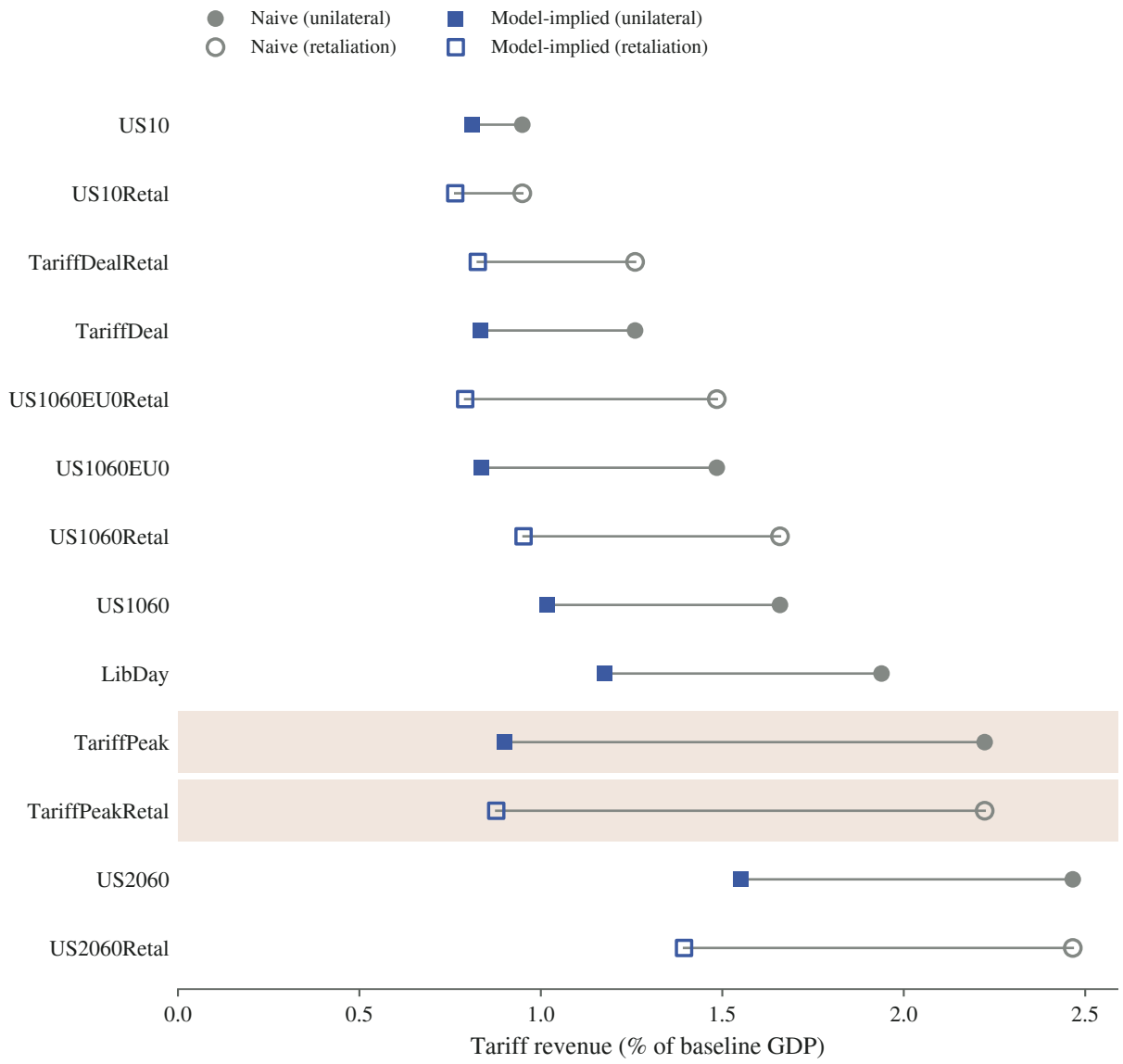
Second, realized revenue is not increasing in the severity of the schedule. Tariff Peak's naive revenue is only slightly smaller than US2060's (2.22% against 2.47%), but the realized revenue is much lower: 0.90% against 1.55%. The same pattern holds for the other scenarios: Tariff Deal, which is more aggressive than Liberation Day, raises less revenue in equilibrium. This is a property of dispersion, not of the level of rates. The mechanism is related to the substitution margin each tariff schedule implies. Under a common elasticity σ across origins, demand for a taxed set of origins falls with an elasticity of approximately $\sigma(1-s)$, where s is the set's expenditure share: the larger the taxed set, the more its price raises the overall price index, and the less buyers can avoid the tax. A uniform tariff taxes all foreign origins at once and leaves relative prices among them unchanged, so buyers can substitute only toward domestic production. Its base erodes slowly: revenue peaks only at a uniform rate of 85%, far above the average rate in any of our scenarios. A dispersed schedule instead concentrates its highest rates on partners that each account for a small share of expenditure, while lightly taxed foreign alternatives remain available. Demand for these partners falls at nearly the full elasticity σ as purchases shift to lower-tariff suppliers. The base therefore shrinks fastest exactly where rates are highest: trade is diverted rather than eliminated, and the rate-weighted base contracts even as statutory rates rise. Dispersion also reallocates market shares on the export side, through a distinct channel: tariffs raise the cost of imported inputs and erode exporters' competitiveness in third markets. Section 6 traces the sourcing effect through this cost channel.

These magnitudes lead to the recurring claim that tariff revenue can substitute for other federal taxation. Even in the most revenue-productive scenario, receipts amount to roughly 14% of total U.S. federal revenue, against around 50% from individual income taxes. Also, the most aggressive tariff schedules are not those that raise the most. Retaliation reduces U.S. revenue in every case despite leaving the U.S. tax schedule unchanged: under US2060, revenue falls from 1.55% to 1.39% of GDP. The channel operates through income, where retaliation depresses U.S. real income and hence import demand, and is absent from partial-equilibrium projections.

The real income gains above assume this revenue is redistributed lump-sum. In Ignatenko et al. (2025) the size of the gain depends on how the revenue is used. It is largest when it offsets a distortionary income tax, and roughly zero under a plain lump-sum rebate. Our model has no distortionary-tax margin, so our baseline already corresponds to their lump-sum case, and the entire gain is a terms-of-trade transfer. Figure 4 isolates how much rests on that transfer by comparing each scenario to the case in which the revenue is instead dissipated.[11] Appendix Section F.2 gives details on its construction. Whenever real income rises

[11] This can be interpreted as revenue that accrues to a narrow set of agents rather than to households at

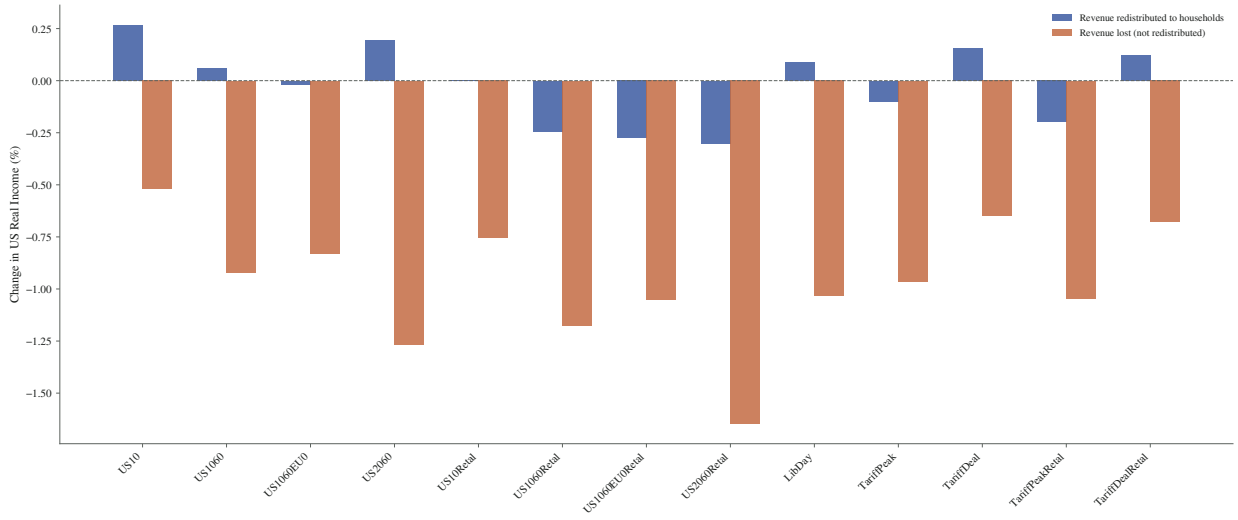
Figure 3: Naive and model-implied U.S. tariff revenue by scenario



Notes: Grey circles report naive tariff revenue, computed by applying the counterfactual tariff schedule to baseline (January 2025) import volumes at baseline prices. Blue squares report tariff revenue in the counterfactual equilibrium, computed from counterfactual final consumption and intermediate flows. Both series are expressed as a share of baseline U.S. GDP; the connecting segment measures the shortfall. Hollow markers denote retaliation scenarios. Scenarios are ordered by naive revenue. Revenue is measured net of the January 2025 applied tariff schedule, which is already embedded in baseline income.

under redistribution, the real income gain reverses once revenue is lost. The gap between the two bars equals counterfactual revenue as a share of baseline income and ranges from about 0.75 to 1.46 percentage points, the largest under US2060. Although small relative to the federal budget, this transfer is several times the large.

Figure 4: No tariff revenues



Notes: Change in U.S. real income relative to the January 2025 baseline, in percent, under the two treatments of tariff revenue. The first bar is the baseline case in which revenue is rebated to the representative household as a lump sum. The second applies an alternative treatment of tariff revenue: counterfactual revenue from the newly imposed schedule is subtracted from U.S. income and welfare is rescaled. Because preferences are homothetic and the upper tier Cobb–Douglas, the correction is exact rather than a first-order approximation, and the pattern of trade and welfare in all other countries is unaffected. The gap between the bars equals counterfactual revenue as a share of baseline U.S. income.

net real income gain in every scenario in which U.S. real income rises. It therefore determines the sign of the change in U.S. real income: once the revenue is lost, the direct price channel from lower foreign producer prices is too weak to offset the distortionary cost of the tariffs.

6 The environmental effects of trade policy

The trade policies studied here were not designed with the climate in mind. Yet by reshaping the geography of global production, they also reshape its carbon content. This section traces that effect and reaches three conclusions. First, the tariffs lower global emissions, but almost entirely by contracting economic activity rather than by making production cleaner. Second, the sourcing effect, which measures whether output moves toward cleaner or dirtier producers, lowers emissions without retaliation and raises them in the retaliatory campaign scenarios. We call the latter *brown sourcing*. Third, the net sourcing effect is small because it is the sum of two large and opposing forces. Buyers in the U.S. market switch to cleaner suppliers, while buyers in all other markets switch to dirtier ones. The headline figure therefore understates how far the tariffs rearrange the carbon content of trade.

6.1 The aggregate effect and its magnitude

Global emissions fall in every scenario, by between around 120.5 and 289.8 Mt CO_2eq (Appendix Figure 15). Retaliation scenarios generate somewhat larger declines than the corresponding unilateral U.S. tariffs. The exception is Tariff Deal, where the two are almost identical. The average reduction across scenarios is comparable to the annual emissions of a mid-sized industrialized economy such as the Netherlands. Valued at the U.S. Environmental Protection Agency's central social cost of carbon of $190 per tonne of CO_2 (2020 dollars) (Rennert et al., 2022; U.S. Environmental Protection Agency, 2023), the reductions correspond to around $22.9 to $55.1 billion per year in avoided climate damages. At the more conservative interim federal value of around $51 per tonne, the figure is still between $6.1 and $14.8 billion per year.

The magnitude is therefore large enough to matter for any welfare accounting of the tariffs. Set against the scale of the climate problem, however, even the largest reduction closes only a small fraction of the gap between current emissions and a Paris-aligned pathway (United Nations Environment Programme, 2024). This shows that trade policy is no substitute for climate policy.

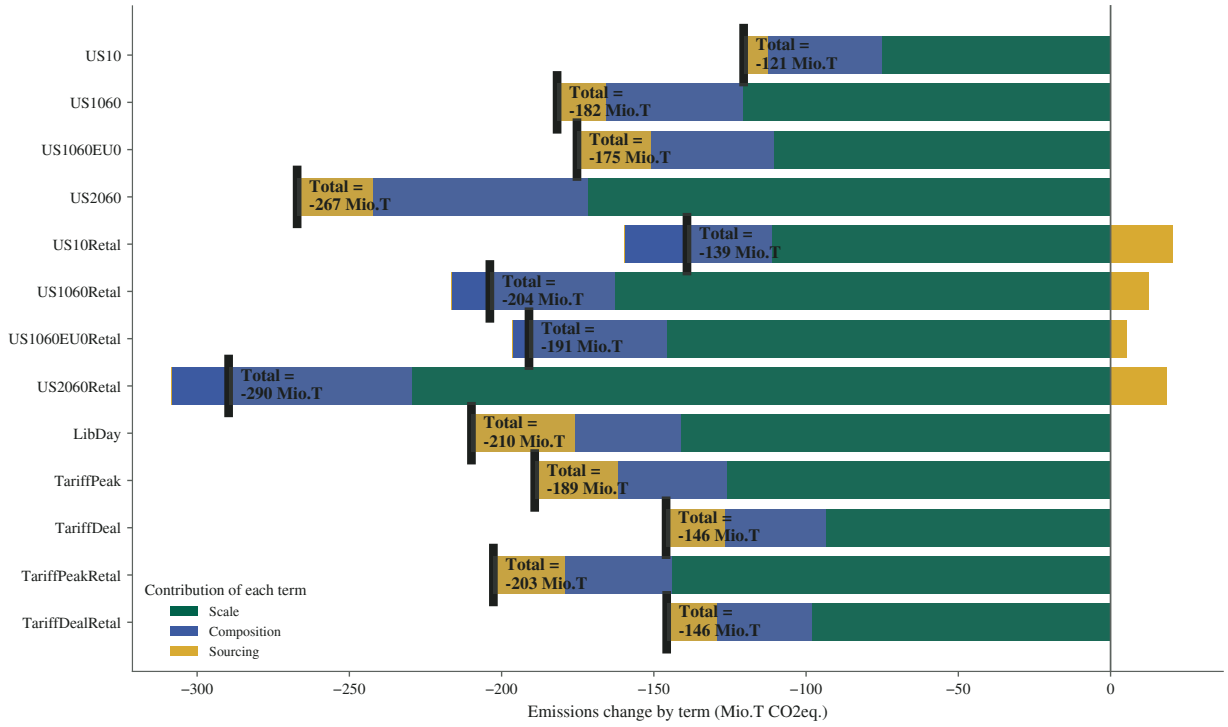
6.2 Decomposing the change: scale, composition, and sourcing

Figure 5 breaks the change in emissions into three channels: scale, composition, and sourcing (see Section 3.3). Normatively these are not equivalent. The scale effect isolates emissions falling because economic activity contracts and is the least desirable mechanism for mitigation, since it lowers emissions only by lowering output. The composition and sourcing effects are preferable: they cut emissions by reallocating activity across sectors or across countries rather than by shrinking the economy. Policies designed to target emissions, such as a uniform global carbon tax (Le Moigne et al., 2024) or a tariff schedule without environmental bias (Shapiro, 2021), work precisely by minimizing the scale channel and amplifying the composition and sourcing channels.

The Trump tariffs do the opposite. Figure 5 shows that the scale effect dominates. It accounts for 62 to 80% of the reduction. The composition effect contributes 17 to 35%. The sourcing effect lowers emissions in 9 of the 13 scenarios, including all five implemented schedules. It raises emissions only in the four retaliatory campaign scenarios (+5.4 to +20.5 Mt). There, the tariffs shift activity away from cleaner producers and toward more carbon-intensive ones, generating brown sourcing.

The tariffs thus reduce global emissions mainly by suppressing economic activity rather than by making trade cleaner. Appendix Figure 16 shows that the scale effect dominates at every level of a uniform U.S. tariff between 5 and 100%, and that without retaliation the sourcing term lowers emissions at every one of these rates. The next subsections show that the small sourcing term hides two large and opposing reallocations,

Figure 5: Global emission changes decomposition



Notes: Contribution of the scale, composition, and sourcing terms to the change in global greenhouse gas emissions in each scenario, in Mt CO_2eq. relative to the January 2025 baseline. Emissions intensities are held at their baseline values, so there is no technique term. Negative contributions lower global emissions and positive contributions raise them. The three terms sum to the total reported to the right of each bar. Two scenarios, TariffPeak and TariffPeakRetal, involve tariffs of up to 145% on China, far above any other rate, and lie outside the range where the first-order decomposition underlying this figure is reliable. Their components should be read with corresponding caution.

and they explain why retaliation tips the balance.

6.3 Decomposition by destination

The sourcing term in Figure 5 measures how far the tariffs shift production toward more or less carbon-intensive suppliers within a given sector. That shift can occur in any market. A U.S. buyer switching suppliers and a Brazilian buyer switching suppliers both contribute. Figure 6 asks *where* the shift takes place. For each scenario, it splits the sourcing term into four parts. The first three parts measure supplier switching within the U.S. market, within the E.U. market, and within all other markets. The fourth part, the destination mix, reflects how the weight of world demand shifts across markets. When world demand shifts toward markets that source more dirtily, average emissions rise even if no buyer changes supplier. The four parts sum to the sourcing term of Figure 5, which is marked separately on each bar.

The split shows that the sourcing term is the net of much larger opposing forces. In the U.S. market, buyers shift toward cleaner suppliers. This lowers emissions by 44.5 Mt CO_2eq under Liberation Day and by 33.9 to 62.3 Mt under the unilateral campaign schedules. It is the market-level counterpart of green protection, part (a) of Proposition 1. In all other markets, sourcing moves in the opposite direction, as predicted by part (b). It raises emissions in every scenario, by 7.2 Mt under Liberation Day and by 8.1 to 23.7 Mt under the unilateral campaign schedules. Supplier switching within the E.U. market is small. It raises emissions in 12 of the 13 scenarios (+1.2 Mt under Liberation Day) and is largest under the E.U. carve-out (+8.7 Mt under US1060EU0). The destination mix raises emissions in 11 of the 13 scenarios (+2.3 Mt under Liberation Day). The exceptions are Tariff Deal and its retaliation case.

Under Liberation Day, cleaner sourcing in the U.S. market outweighs the three emissions-increasing parts combined. This leaves a net sourcing term of −33.9 Mt. The net lowers emissions in 9 of the 13 scenarios. It turns positive only in the four retaliatory campaign scenarios (+5.4 to +20.5 Mt). There, dirtier sourcing in other markets rises to between 35.8 and 63.3 Mt, and the destination mix adds a further 16.9 to 33.7 Mt. The sourcing channel is therefore not second-order. Its net is small because large opposing flows partly cancel, not because the tariffs barely rearrange the carbon content of global production.

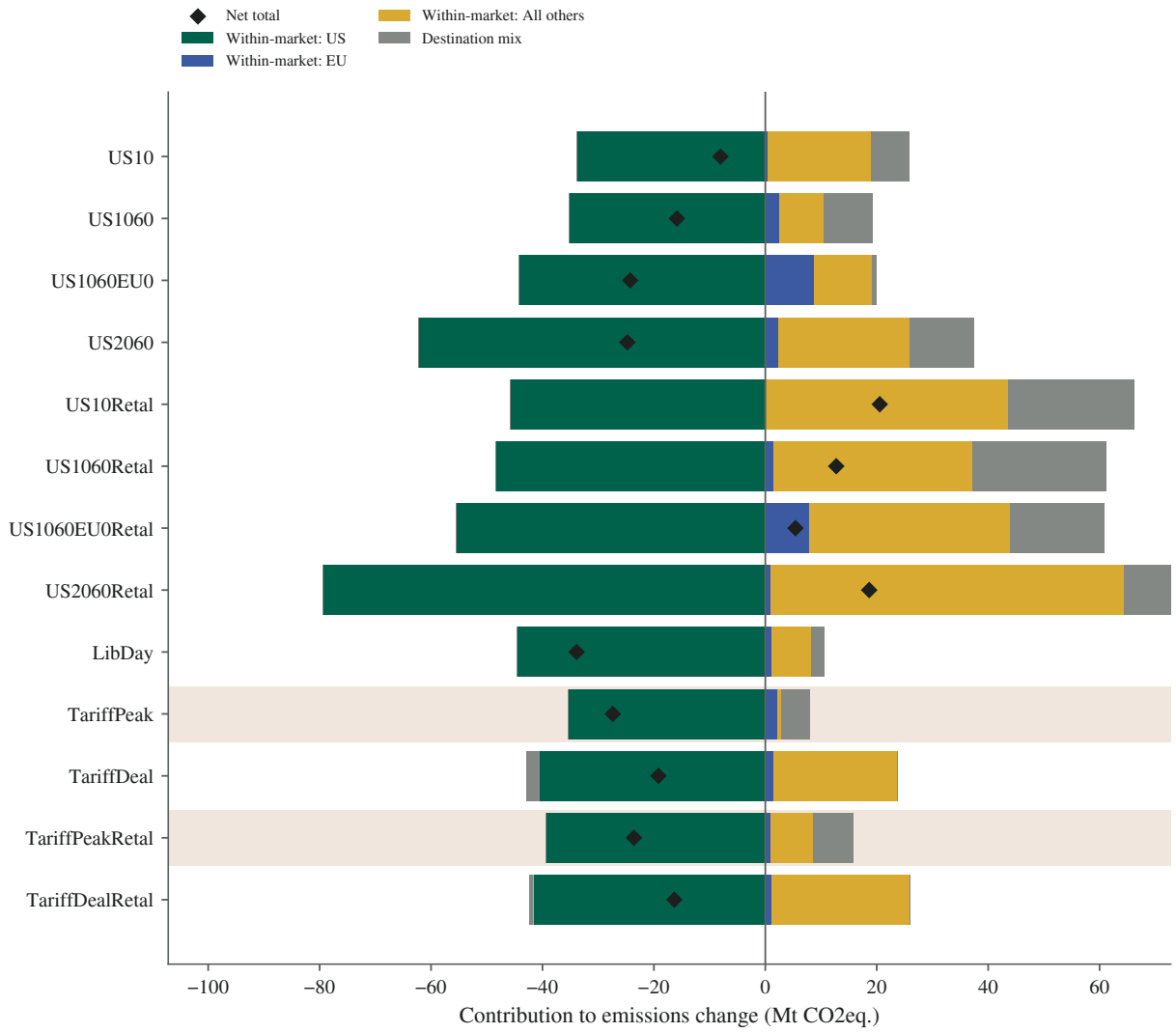
6.4 Mechanism: the dirty-origin share in U.S. and abroad

Figure 7 tests the two predictions of Proposition 1 directly. For each scenario it reports the change in the within-sector market share of carbon-intensive origins, separately for sales into the United States and for sales into all other markets. The proposition predicts that this share falls in the tariff-imposing market and rises in third markets.

Both predictions hold in every scenario. The market share of dirty origins in the U.S. market falls by between 0.12 and 0.70 percentage points. This is green protection: the tariffs shift U.S. purchases toward cleaner suppliers. Third markets instead experience brown sourcing. Their dirty-origin share rises by 0.11 to 0.24 percentage points in the unilateral scenarios and by up to 0.29 percentage points in the retaliation scenarios. As predicted by Corollary 1, retaliation strengthens it. Under US2060, for example, the increase is 0.24 percentage points without retaliation and 0.29 with it. The third-market shifts are small in raw share terms. They matter for emissions because they are concentrated in the most carbon-intensive sectors, which carry more weight once shares are weighted by emission intensity, as in the sourcing term.

Parts (a) and (b) of Proposition 1 hold for any tariff. Which of the two forces dominates in the aggregate is a separate question, and condition (17) answers it. For a large clean economy such as the United States, the pass-through of the tariff into a country's factory-gate price Λ in equation (15) is small. It is around

Figure 6: Sourcing effect decomposition



Notes: Decomposition of the sourcing term by destination market, in Mt CO_2eq. The three within-market components hold each region's baseline market size fixed and measure how far origin shares shift within the U.S. market, within the E.U. market, and within all remaining destinations. The destination-mix component holds every origin's within-market shares fixed and measures the effect of the changing relative size of markets, which raises average emissions when demand shifts toward markets that source more carbon-intensively. The four components sum exactly to the aggregate sourcing term of Figure 5, marked separately on each bar. Positive values raise global emissions. Two scenarios, TariffPeak and TariffPeakRetal, involve tariffs of up to 145% on China, far above any other rate, and lie outside the range where the first-order decomposition underlying this figure is reliable. Their components should be read with corresponding caution.

0.021, just below the threshold of around 0.028. The condition therefore predicts that unilateral tariffs leave the within-market reallocation on the green side. The data agree with this prediction. The within-market part of the sourcing term, the sum of the U.S., E.U., and other-market parts in Figure 6, is negative in all 13 scenarios. It ranges from −36.2 Mt (US2060 and Liberation Day) to −2.1 Mt (US10Retal). Retaliation

pushes it toward zero, as Corollary 1 predicts, but does not reverse its sign. Under US10 it is −15.0 Mt without retaliation and −2.1 Mt with it. Under US2060 it is −36.2 and −15.0 Mt.

Retaliation turns the net sourcing term positive because the destination mix then outweighs the within-market part.[12] The destination mix raises emissions in 11 of the 13 scenarios. Destinations that source more carbon-intensively than the world average, such as the rest of the world, Russia, and Saudi Arabia, contribute positively in every scenario. Together, all destinations other than the United States add between 2.6 and 18.2 Mt, with little difference between scenarios with and without retaliation. What changes with retaliation is the contribution of the United States, which sources around 9% less carbon-intensively than the world in the same sectors. In 9 of the 13 scenarios, including the retaliation cases of Tariff Peak and Tariff Deal, the U.S. share of world absorption rises slightly (by 0.02 to 0.07 percentage points), and the U.S. contribution to the destination mix is negative (between −13.1 Mt and around zero). In the four retaliatory campaign scenarios, the U.S. share falls (by 0.05 to 0.08 percentage points), and the U.S. contribution turns positive (+10.8 to +15.5 Mt). Relative to the corresponding unilateral scenario, retaliation thus raises the destination mix by 15.2 to 22.2 Mt. It also shrinks the within-market part by a similar amount (12.9 to 21.2 Mt), while staying negative. The positive net therefore results from the destination mix.

6.5 Mechanism: the cost channel

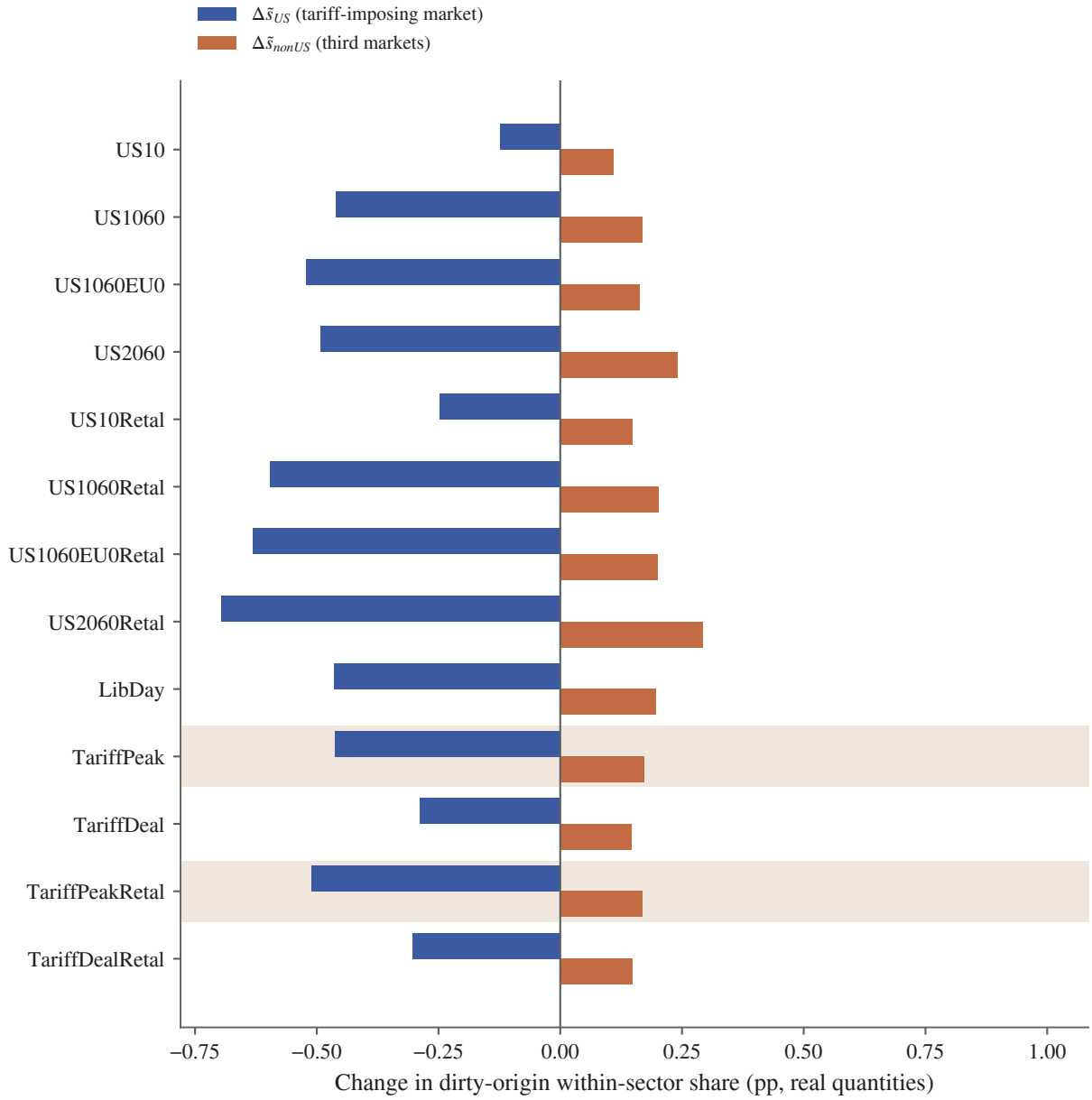
The dirty-share results show that third-market sourcing shifts, but not that the shift runs through input costs, as the proposition claims. To isolate the cost channel, we use a first-order approximation of the within-market part of the sourcing term. For destination n and sector s, CES demand implies that the within-market change in emissions is approximately $-\varepsilon_s X_{ns} \operatorname{Cov}_\pi(e_{is}, \hat{c}_{ins})$. Here ε_s is the trade elasticity of sector s, X_{ns} is destination n's absorption of sector s, and the covariance is taken across origins i, weighted by their market shares π.[13] In intuitive terms, emissions fall when the tariffs make dirty origins more expensive relative to clean ones, and rise when clean origins become relatively more expensive. The expression tracks the within-market part closely. Across the 13 scenarios the correlation is 0.98 and the sign agrees in 13 of 13 cases.

The delivered-cost change has two components: the direct tariff wedge and the change in the producer price. The second component contains the higher cost of tariffed inputs. Splitting the covariance accordingly

[12] The proposition holds market sizes fixed, so it does not speak to the destination mix.

[13] The shares π and the absorption X_{ns} are midpoints of their baseline and counterfactual values, as in the within-market part of Figure 6. The expression is therefore a decomposition of the model solution rather than an ex-ante prediction. Final and intermediate purchases are pooled. The delivered-cost change is $\hat{c}_{ins} = \ln \hat{p}_{is} + \ln[(1 + t^0_{ins} + \tau_{ins})/(1 + t^0_{ins})]$, with t^0 the January 2025 tariff and τ the scenario tariff. Trade costs are constant in the model, so they do not contribute.

Figure 7: Mechanism, Proposition 1: dirty-origin share, U.S. versus third markets



Notes: Change in the within-sector quantity share of carbon-intensive origins, in percentage points, computed separately for sales into the United States and for sales into all other destinations. An origin is classified as carbon-intensive if its baseline emission intensity, pooled across all sectors, exceeds the unweighted mean across countries. The classification is therefore the same in every sector. Shares are aggregated across sectors using baseline sales weights. Two scenarios, TariffPeak and TariffPeakRetal, involve tariffs of up to 145% on China, far above any other rate, and lie outside the range where the first-order approximation underlying this figure is reliable. Their values should be read with corresponding caution.

isolates the cost channel. The direct-tariff part lowers emissions in all 13 scenarios, by between 16.0 and 109.8 Mt, because the tariffs fall more heavily on carbon-intensive origins. This is the green protection

mechanism. The induced-cost part raises emissions in all 13 scenarios, by between 10.6 and 71.4 Mt, because producer prices rise more for cleaner origins. This is the cost channel of Proposition 1. Under Liberation Day, the direct-tariff part is −81.0 Mt and the induced-cost part +43.3 Mt. In the retaliatory campaign scenarios, the direct-tariff part is much smaller (−16.0 to −35.7 Mt), since the retaliatory tariffs fall on U.S. exporters, which are relatively clean.

6.6 Territorial vs. footprint emissions

The emissions changes reported so far are territorial, meaning they record where emissions are physically released. Since tariffs act mainly on trade, a territorial accounting attributes the entire adjustment to producing countries and says nothing about whose consumption the emissions serve. Figure 8 therefore reports, for Liberation Day, both the change in territorial emissions and the change in each country's carbon footprint. The footprint is the emissions released anywhere in the world to satisfy a country's final demand, at home or abroad, directly or embodied in intermediate inputs along the supply chain. Appendix Section H explains the construction. Exports enter the territorial measure but not the footprint, and imports enter the footprint but not the territorial measure. The difference between the two therefore isolates the international redistribution of embodied emissions.

The first pattern concerns the United States. U.S. territorial emissions fall by around 11.7 Mt CO_2eq, while the emissions released anywhere in the world to serve U.S. final demand fall by around 199.8 Mt, a factor of around 17.1 more. This fall in the U.S. footprint does not come from lower U.S. consumption. U.S. real final consumption rises slightly under Liberation Day (by 0.33%), so the scale part of the U.S. footprint is positive (+19.4 Mt). The fall comes almost entirely from sourcing (−210.3 Mt, Appendix Figure 13). Around half of it (−108.6 Mt) reflects U.S. consumers switching to cleaner origins. The other half (−101.7 Mt) reflects the suppliers that U.S. consumers already buy from becoming less carbon-intensive, as these suppliers in turn source their inputs from cleaner origins. This is green protection seen from the consumption side. Since much of the avoided emissions would have been released abroad, a territorial accounting assigns them to the exporting countries. China shows the reverse pattern on a smaller scale. Its territorial emissions fall by around 40.7 Mt against a footprint decline of around 36.0 Mt. The emissions released on Chinese soil thus fall by more than those released to serve Chinese final demand.

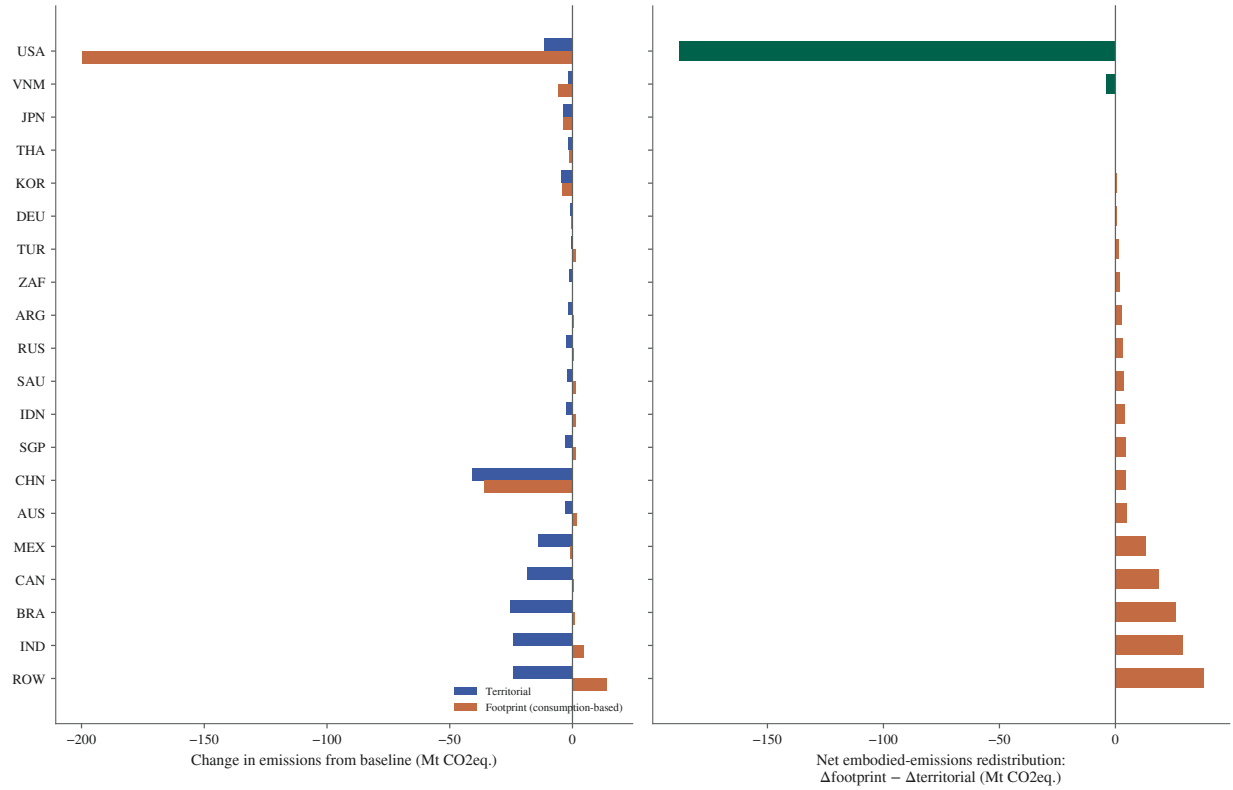
The second and more substantive pattern concerns third countries. Several economies register an increase in consumption-based emissions even as their territorial emissions fall, most visibly the rest-of-world aggregate, India, Canada, Indonesia and Australia. Because emissions embodied in exports do not enter the footprint measure, such an increase cannot be generated by the contraction in foreign demand for these

countries' output. It requires that their own final demand has become more carbon-intensive. They source a larger share of what they consume from more emission-intensive producers, either directly or through their supply chains.

This is the consumption-side counterpart of the brown sourcing in third markets documented in Sections 6.4 and 6.5. As the tariffs raise the delivered cost of goods from relatively clean, trade-exposed suppliers, third markets substitute toward dirtier origins. In these countries the shift is large enough to outweigh the fall in the overall scale of activity. On the production side, the dirtier producers that these markets source from gain relative to others, but not in absolute terms. None of the twenty largest emitters records a rise in territorial emissions under Liberation Day (Russia's, for example, fall by 2.6 Mt), because the global demand contraction lowers production almost everywhere. A country-level decomposition of the footprint change confirms this (Appendix Section I). Under Liberation Day, the footprint rises in 51 economies, by 47.0 Mt in total. Sourcing adds 67.1 Mt, while lower consumption subtracts 19.6 Mt and the sectoral composition of final demand barely matters (−0.5 Mt). Of the sourcing part, 39.4 Mt comes from these economies buying from dirtier origins and 27.6 Mt from their suppliers' own inputs becoming more carbon-intensive. The same pattern holds in every scenario (Appendix Figure 14). The sourcing part of the U.S. footprint is negative in all 13 scenarios, between −339.2 and −111.6 Mt. Of the 19 other economies in the figure, 16 have a positive sourcing part in every scenario, among them the rest of the world, India, Mexico, and Canada.

The right panel of Figure 8 reports the difference between each country's footprint and territorial changes. A positive value means that a country's net imports of embodied emissions rise. This difference should be read with care. It rises either when a country absorbs more embodied emissions or when it supplies fewer to others, and the contraction in U.S. import demand raises it mechanically for every large exporter. Brazil's net embodied emissions for example rise by 26.1 Mt almost entirely because its territorial emissions fall (by 25.3 Mt), while its footprint barely moves (+0.8 Mt). The evidence for dirtier sourcing therefore rests on the rise in the footprint itself, not on a country's position in the right panel. Nor does this evidence depend on the ICIO rest-of-world aggregate. In every scenario, the other economies with rising footprints and falling territorial emissions together account for a larger footprint increase than the rest of the world, by a clear margin. Under the E.U.-inclusive schedules, this group includes several major European manufacturers. Appendix Figure 17 reports the gap across all scenarios and shows it to be stable.

Figure 8: Territorial and consumption-based emissions changes under the Liberation Day tariff schedule



Notes: Changes in greenhouse gas emissions relative to the January 2025 baseline under the Liberation Day schedule, in Mt CO_2eq. Territorial emissions are those physically released within a country, including emissions generated in producing exports. Consumption-based emissions (the carbon footprint) are those released anywhere in the world to serve a country's final demand, excluding emissions embodied in exports and including emissions embodied in imports. The right panel reports the difference between the two; a positive value indicates a rise in a country's net imports of embodied emissions. This difference is a net magnitude and improves either when a country absorbs more embodied emissions or when it supplies fewer to others, so evidence of dirtier sourcing rests on the footprint change itself rather than on a country's position in the right panel. The plotted economies are the twenty largest baseline emitters by territorial emissions plus the ICIO rest-of-world region; economies outside this window are omitted, so the bars do not sum to the global total.

7 Robustness

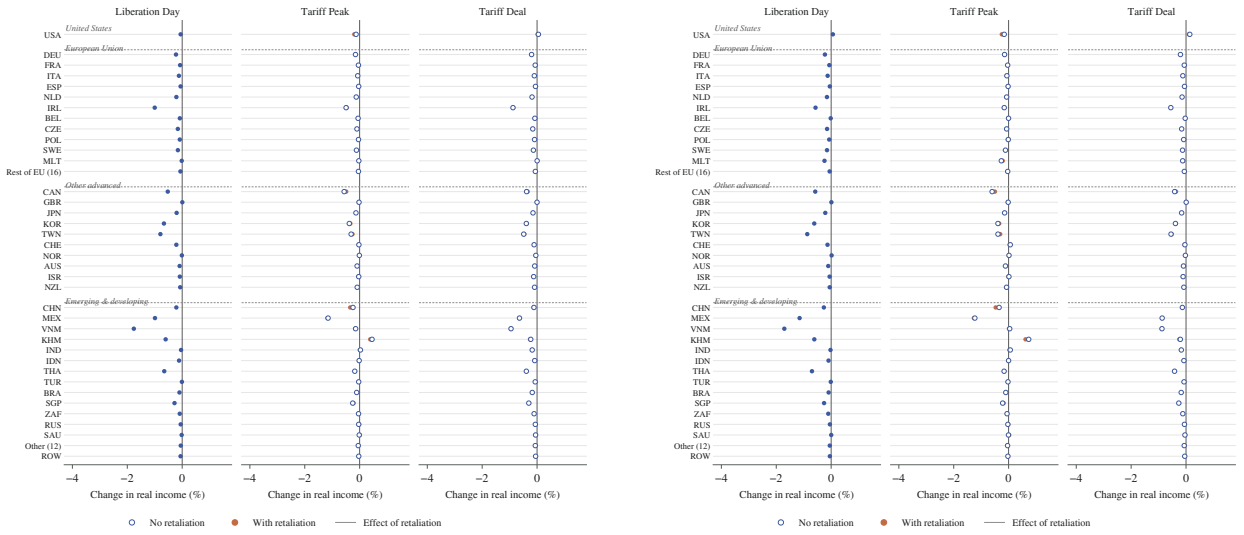
The trade elasticities govern how strongly buyers substitute between origins. They therefore matter most for the sourcing term, which measures exactly this substitution. This section asks how far our results depend on them. Our main analysis uses elasticities estimated with the method of Caliendo and Parro (2015). We first solve all scenarios under two alternative sets. The first applies the method of Fontagné et al. (2022) to our data, supplemented with their tariff data. The second takes the elasticities of Fontagné et al. (2022) directly and aggregates them to our sectors. Appendix G reports all three sets.

Figure 9 shows the real income changes under the implemented schedules for the two alternative elasticity sets. The choice of elasticities changes little. Both the pattern across countries and the magnitudes are close to those in Figure 2.

Figure 10 decomposes the change in global emissions for the two alternative sets, constructed as in Figure 5. The scale effect dominates under all three sets. The total decline in emissions is not systematically larger under the Fontagné et al. (2022) elasticities, although they are the highest of the three elasticity sets (mean 5.07, against 3.61 for Caliendo and Parro (2015) and 4.00 for the aggregated set). The decline is larger than under the Caliendo and Parro (2015) set in 6 of the 13 scenarios and smaller in the others, including all four retaliatory campaign scenarios.

The sourcing term has the same sign under all three sets in all 13 scenarios, but its size differs. Under Liberation Day it is −66.3 Mt under the Fontagné et al. (2022) elasticities, against −33.9 Mt under Caliendo and Parro (2015) and −19.8 Mt under the aggregated set. In all 13 scenarios, the term is lowest under the Fontagné et al. (2022) elasticities and highest under the aggregated set. Most of this ordering comes from one sector, agriculture. In all three sets, agricultural reallocation lowers emissions, and a larger elasticity amplifies this effect. The agricultural elasticity estimated with the Fontagné et al. (2022) method is around 67% higher than the one estimated with the Caliendo and Parro (2015) method and 2.86 times the aggregated one. Basic metals partly offsets this, since its reallocation raises emissions in most scenarios. Its elasticity, however, differs less across the sets. It is similar under the Caliendo and Parro (2015) and the aggregated sets and 23% higher under the Fontagné et al. (2022) method. The parts of the sourcing term also keep their signs. Under all three sets and in all 13 scenarios, the U.S.-market part lowers emissions and the part for all other markets raises them. The dirty-origin share falls in the U.S. market and rises in third markets. The within-market part as a whole is negative in every scenario under the first two sets. The exception is US10Retal under the aggregated set, where it is +0.6 Mt.

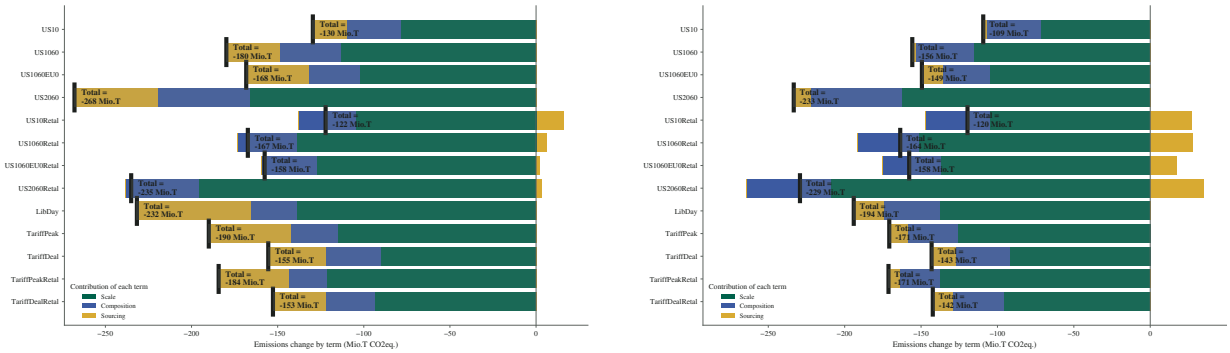
Figure 9: Real income changes Implemented scenarios under alternative elasticity estimates



(a) Fontagné et al. estimation (b) Aggregated elasticities

Notes: Change in real income relative to the January 2025 baseline, in percent, under the implemented schedules, constructed exactly as in Figure 2 but solving the model with alternative trade elasticities. Panel (a) uses elasticities estimated on our sample with the method of Fontagné et al. (2022). Panel (b) uses the elasticities of Fontagné et al. (2022) aggregated to our sectoral definitions. The baseline figure uses the Caliendo and Parro (2015) estimates. All three sets are reported in Appendix Table 3.

Figure 10: Global emission changes decomposition under alternative elasticity estimates



(a) Fontagné et al. estimation (b) Aggregated elasticities

Notes: Contribution of the scale, composition, and sourcing terms to the change in global emissions, in Mt CO_2eq., under alternative trade elasticities. Panel (a) uses elasticities estimated on our sample with the method of Fontagné et al. (2022). Panel (b) uses the elasticities of Fontagné et al. (2022) aggregated to our sectoral definitions. Construction is otherwise identical to Figure 5, which uses the Caliendo and Parro (2015) estimates. The caution regarding TariffPeak and TariffPeakRetal noted in Figure 5 applies here as well.

8 Conclusion

This paper asks whether protectionism by a relatively clean economy makes global production cleaner. We answer the question with the U.S. tariffs of 2025, which we evaluate in a quantitative trade model with global input-output linkages. The tariffs lower global emissions in every scenario, by between 120.5 and 289.8 Mt CO_2eq. Most of this reduction comes from lower economic activity. The scale effect alone accounts for 62 to 80% of it. Tariffs thus reduce emissions in the most expensive way available, by making the world poorer.

The margin on which trade policy could be the right climate instrument is sourcing, since discrimination by origin is the one thing a domestic instrument cannot do. On this margin, the optimistic argument holds in the net for the tariffs as applied, but for a fragile reason. A tariff on dirty origins makes the protected market source more cleanly. At the same time, it raises the input costs of the clean country and erodes its competitiveness in every other market, where dirtier producers gain ground. We show this trade-off analytically and find both forces in every one of our thirteen scenarios. Under Liberation Day, cleaner sourcing in the U.S. market outweighs dirtier sourcing everywhere else. When partners retaliate across the board, the U.S. market shrinks relative to the rest of the world, and the net sourcing effect turns brown. The net change in global emissions therefore hides how far tariffs rearrange the carbon content of trade.

These results have two implications for policy. First, tariffs are no substitute for climate policy. Even the largest reduction we find closes only a small fraction of the gap between current emissions and a Paris-aligned pathway, and it comes at the cost of lower real income in most countries. Second, the environmental case for protection depends on the response of trading partners. A clean economy that raises tariffs exports brown sourcing to third markets, and retaliation strengthens it.

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A Data Sources and Coverage

Our data sources follow the approach of Le Moigne et al. (2024), integrating trade and emissions data from multiple international databases to ensure broad coverage of global economic activity.

Trade Data: We use the OECD Inter-Country Input-Output (ICIO) tables (OECD, 2023) for trade flow data, which provide a detailed representation of global supply chains. The ICIO tables report both intermediate and final goods trade across 42 industries and 64 economies, including an aggregated Rest of the World category. The dataset spans 1995–2018, with 2018 serving as the baseline year for model calibration, while earlier years are used to estimate key elasticities.

Tariff Data: We obtain bilateral applied import tariffs for 2018 from the World Bank (WITS). On top of these, we add the trade-war measures in force in January 2025 and the 2025 U.S. tariff schedules, which we build at the product level. Appendix C summarizes the construction, and the Online Appendix documents it in full.

Emissions Data: We compile greenhouse gas (GHG) emissions data, expressed in CO_2 equivalents, from three complementary sources: (i) OECD's TECO2 dataset (OECD, 2021), which reports CO_2 emissions from fuel combustion across the same 42 industries and 64 economies covered in the ICIO tables; (ii) FAOSTAT Emissions Totals dataset (FAO, 2023), which accounts for agriculture, forestry, and land use emissions; (iii) European Commission's EDGAR database (European Commission, 2023), which includes industrial processes, product use, and fugitive emissions.

By integrating these sources, we capture approximately 93% of global GHG emissions, ensuring a comprehensive dataset for analyzing trade-related emissions.

While our data provides extensive coverage of global production, trade, and emissions, it has two key limitations: First, our emissions intensity calculations rely on trade values rather than physical trade volumes. This is due to the ICIO tables reporting trade in monetary terms, with no suitable price deflators available at this level of aggregation. Consequently, in our counterfactual analysis, changes in emissions reflect both shifts in trade volumes and price fluctuations.

Second, we assume that a ton of steel produced in Germany has the same emissions intensity whether it is consumed domestically or exported to the U.S. While this assumption simplifies the analysis, it aligns with the iceberg cost formulation of transport emissions in our model. That is, a ton of German steel delivered to

the U.S. effectively carries higher embodied emissions due to transportation losses (e.g., 20% "melting away" in transit). However, because these losses also increase the price of imported goods, the emissions intensity per unit of trade remains unchanged in our framework.

B Data treatment

Aggregations

To avoid sparseness of the input-output table and zero gross outputs, we aggregate the following economies:

- Luxembourg and Belgium: subsequently labeled `BEL` in all data
- Hong-Kong, China and China: subsequently labeled `CHN`
- Malaysia and Singapore: subsequently labeled `SGP`

as well as the following sectors:

- 'Mining and quarrying, energy producing products' [D05T06] with 'Mining support service activities' [D09]: subsequently labeled as [D05T06] (Mining, energy)
- 'Motor vehicles, trailers and semi-trailers' [D29] with 'Other transport equipment' [D30]: subsequently labeled [D29T30] (Transport equipment)

These aggregations leave us with a sample of 64 economies (incl. ROW aggregate) and 42 sectors from 1995 to 2018.

ICIO

The raw ICIO tables records negative values for some accounts of final consumption or value added. As the model cannot accommodate these negative values, we redistribute the negative parts in the table while respecting the following constraints:

- the sum of the columns and the sum of the rows must remain equal,
- the technical coefficients within the IO table (intermediate input spending over gross output ratio, corresponding to the parameters γ in the model) must remain constant equal to the raw ratios.

FAO

We keep only FAO Tier 1 emissions by subcategories belonging to the category 'Agricultural Land' with the exception of 'On-farm Energy Use', since these emissions are already contained in the TECO2 emission

data.[14] The remaining observations are then aggregated into the 64 economies with the ROW aggregate and are assigned to the 'Agriculture' sector.

EDGAR

We first combine different time series extracts of the EDGAR database, namely the 'CH4', 'CO2_excl_short-cycle_org_C' and 'N2O' data sheet by converting the emissions into CO_2 equivalents according to the respective AR4 100-year GWP value.[15] We then aggregate the data into our 63 sample economies and create the ROW aggregate with the remaining economies. To assign the IPCC emission categories to our various sample sectors, we rely on the exact definition of the IPCC emission category compared to the ISIC rev.4 codes comprised in our sample sector definition.

For IPPC category 'industrial process and product use emissions' (chapter 2), we apply the following conversion:

IPCC category	Name	Sample sector
2.A	Mineral Industry	Non-metallic minerals
2.B	Chemical Industry	Chemicals
2.C	Metal Industry	Basic metals
2.E	Electronics Industry	Electronic
2.F	Product Uses As Substitutes For Ozone Depleting Substances	Energy

For the IPCC categories "fugitive emissions" (chapter 1.B) we proceed in two steps. Based on the categories definitions we have a direct mapping for the subcategory 'Oil and Natural Gas' (1.B.2) assigned to the sample sector 'Mining, energy'. The subcategory 'Solid Fuels' (1.B.1) however matches with different sample sectors: 'Mining, energy', 'Mining, non-energy', 'Wood', and 'Coke, petroleum'. We therefore disaggregate the IPCC aggregate "Solid fuelds" into the respective sample sectors by using as a disaggregation weights the share of emissions from fuel burning of each sample sector in the total.[16]

[14] The category 'Agricultural Land' includes the following subcategories: 'Fires in humid tropical forests', 'Fires in organic soils', 'Net Forest conversion', 'Drained organic soils', 'Synthetic Fertilizers', 'Crop Residues', 'Manure left on Pasture', 'Manure applied to Soils', 'Manure Management', 'Enteric Fermentation', 'Savanna fires', 'Burning - Crop residues', 'Rice Cultivation', 'On-farm Energy Use'

[15] The AR4 100-year GWP values are 25 for CH_4 and 298 for N_2O.

[16] Note that we did not include the IPCC categories 2.D 'Non-Energy Products From Fuels and Solvent Use' and 2.G 'Other Product Manufacture and Use' since a clean mapping from the IPCC categories to the corresponding sample sectors is not as easily separable.

C Tariff data

This appendix summarizes how we build the tariffs used in the model. The Online Appendix documents every measure, every rule we apply, and every approximation we make. The tariffs have three layers. The first layer is the 2018 baseline: the applied MFN and preferential tariffs of 2018 from WITS, under which the ICIO 2018 trade flows are observed. The second layer is the January 2025 status quo. It adds to the 2018 baseline the trade-war measures that were still in force on 1 January 2025 and that the WITS data do not contain. These are the Section 301 duties on China, the Section 232 duties on steel and aluminum, the solar safeguard, the column 2 rates on Russia, and the retaliation by China, Russia, and Turkey. The third layer is the 2025 scenarios, each of which is the full US tariff schedule on one date in 2025, with and without foreign retaliation. We solve the status quo as a counterfactual relative to 2018 and use the resulting equilibrium as the baseline of the paper. Every scenario, including the campaign scenarios, is then solved as an increment on top of the status quo. All effects in the paper are therefore effects relative to the trade-war world of January 2025, not relative to 2018.

We determine all tariffs at the product level first and aggregate them to the model's sectors afterwards. A product is a 2018 ten-digit HTS line imported from a given partner or, for retaliation, exported by the United States to a given partner. The trade values come from the replication files of Fajgelbaum et al. (2020a). The tariff of a partner in a sector is the average over the products in that sector, weighted by 2018 trade. Since the weights are the same for every layer, differences between the status quo and the scenarios come only from tariff rates. The status quo measures are read from the 2025 Basic Edition of the Harmonized Tariff Schedule (HTSUS). Each 2025 scenario uses the HTSUS revision in force on its date: 9 April 2025 for LibDay, 12 May 2025 for TariffPeak, and 17 November 2025 for TariffDeal. Where several measures apply to the same product, we combine them line by line following the executive orders. For example, goods under a Section 232 duty are exempt from the reciprocal tariff, and goods from Canada and Mexico that claim USMCA preference are exempt from the fentanyl duty.

Table 2 in Section 2 reports the resulting tariffs by partner. In the status quo, US goods imports pay an additional 3% on average, almost all of it on China. On 2018 trade weights, our averages are close to the trade-weighted averages of Bown (2025) at the peak of the US–China escalation and somewhat lower at the other dates. The Online Appendix reports these averages and shows that most of the gap comes from the trade weights and from the share of Canadian and Mexican imports that claims USMCA preference.

D Model - Equilibrium in Changes

This section describes the model equilibrium in changes using Dekle et al. (2007)'s "exact hat algebra". This involves re-writing variables as linear changes from the baseline. In what follows, a baseline version of a variable x is denoted by x^B. The proportional change is then given by $\tilde{x} = x/x^B$.

Following this procedure, changes in the demand for final goods, the demand for inputs, price indexes, and ex-factory prices are given by:

$$\widetilde{q}_{is'j} = \widetilde{I}_{s'j}[\widetilde{p}_{is'}\frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{-\sigma_{s'}} \widetilde{P}^{c\ (\sigma_{s'}-1)}_{s'j} \tag{19}$$

$$\widetilde{P}^{c}_{s'j} = \left(\sum_i [\widetilde{p}_{is'}\frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{(1-\sigma_{s'})}\left(\frac{q^B_{is'j}p^B_{is'j}(1+t^B_{is'j})}{I^B_{s'j}}\right)\right)^{\frac{1}{(1-\sigma_{s'})}} \tag{20}$$

$$\widetilde{m}_{is'js} = \widetilde{E}_{s'js}[\widetilde{p}_{is'}\frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{-\eta_{s's}} \widetilde{P}^{p\ (\eta_{s's}-1)}_{s'js} \tag{21}$$

$$\widetilde{P}^{p}_{s'js} = \left(\sum_i [\widetilde{p}_{is'j}\frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{(1-\eta_{s's})}\left(\frac{m^B_{is'j}p^B_{is'js}(1+t^B_{is'j})}{E^B_{s'js}}\right)\right)^{\frac{1}{(1-\eta_{s's})}} \tag{22}$$

$$\widetilde{p}_{js} = \widetilde{w}_j^{\gamma_{j,Ls}} \prod_{s'} \widetilde{P}^{p\ \gamma_{s'js}}_{s'js} \tag{23}$$

Changes in the market clearing conditions are given by:

$$\widetilde{I}_{s'j} = \sum_s \widetilde{w}_j \widetilde{L}_{js}(w^B_j L^B_{js}) + \sum_{i,s',s} \widetilde{p}_{is'}\widetilde{m}_{ijs's}t_{ijs'}(p^B_{is'j}m^B_{is'js}) + D^B_j \tag{24}$$

$$\widetilde{E}_{js} = \sum_i \left(\widetilde{p}_{js}\widetilde{q}_{jsi}(p^B_{jsi}q^B_{jsi}) + \sum_{s'} \widetilde{p}_{js}\widetilde{m}_{jsis'}(p^B_{jsi}m^B_{jsis'})\right) \tag{25}$$

$$\sum_s \frac{\widetilde{E}_{js}}{\widetilde{w}_j} L^B_{js} = L_j \tag{26}$$

E Proof of Proposition 1

This appendix proves Proposition 1 and Corollary 1, and establishes the two extensions discussed at the end of Section 3.4. Throughout, all derivatives are evaluated at $t = 0$, wages are held fixed, and the final demand quantity indices X_C and X_D are held fixed.

Step 1: Hicksian demand

Conditional on attaining X_j at minimum cost, expenditure minimisation over the CES aggregate delivers

$$q_{ij} = \theta_{ij} \left(\frac{p_i^j}{P_j} \right)^{-\sigma} X_j, \qquad P_j = \left[\sum_i \theta_{ij} \, (p_i^j)^{1-\sigma} \right]^{1/(1-\sigma)}, \tag{27}$$

where p_i^j is the delivered price of origin i's variety in market j. This is the compensated counterpart of equation (3), obtained by replacing the income term $\beta_{s'j} I_j$ with the fixed quantity target X_j. Consumer and producer prices are linked by

$$p_C^C = p_C, \qquad p_D^C = \tau_{DC} \, p_D \, (1+t), \qquad p_C^D = \tau_{CD} \, p_C, \qquad p_D^D = p_D,$$

so the tariff drives a wedge only in market C, while the response of p_C propagates into market D through unit costs.

Step 2: Cost pass-through

Under perfect competition producer prices equal unit costs. With Cobb-Douglas technology over labour and the intermediate bundle,

$$\ln p_i = \text{const} + (1-\alpha) \ln w_i + \alpha \ln P_{M,i}, \qquad P_{M,C}^{1-\eta} = \omega_{CC} \, p_C^{1-\eta} + \omega_{DC} \big(\tau_{DC} p_D (1+t) \big)^{1-\eta}.$$

Differentiating the CES price index and using $\bar{\omega}_{kC}$ for the baseline cost shares gives $d \ln P_{M,C} = \bar{\omega}_{CC} \, d \ln p_C + \bar{\omega}_{DC} (d \ln p_D + dt)$; the elasticity η drops out because at $t = 0$ the index is evaluated at the cost-minimising bundle. With $\omega_{CD} = 0$, D's unit cost depends only on w_D and p_D, so $d \ln p_D = \alpha \, d \ln p_D$ and hence $d \ln p_D = 0$. Substituting and holding w_C fixed,

$$d \ln p_C = \alpha \big[\bar{\omega}_{CC} \, d \ln p_C + \bar{\omega}_{DC} \, dt \big],$$

which is a fixed point in $d \ln p_C$ because C's own output enters its intermediate bundle. Solving,

$$\left. \frac{\partial \ln p_C}{\partial t} \right|_{t=0} = \frac{\alpha \bar{\omega}_{DC}}{1 - \alpha \bar{\omega}_{CC}} \equiv \Lambda.$$

Since $1 - \alpha \bar{\omega}_{CC} = 1 - \alpha + \alpha \bar{\omega}_{DC} > \alpha \bar{\omega}_{DC} > 0$ for $\alpha < 1$, we have $\Lambda \in (0, 1)$, which establishes equation (15). Omitting the fixed point and reading $\partial \ln p_C / \partial t = \alpha \bar{\omega}_{DC}$ directly from the price index understates the

pass-through by the Domar multiplier $(1-\alpha\bar{\omega}_{CC})^{-1}$.

Step 3: Relative-price responses

In market C, $\ln(p_D^C/p_C^C) = \ln\tau_{DC} + \ln(1+t) + \ln p_D - \ln p_C$, so

$$\frac{\partial}{\partial t}\ln\left(\frac{p_D^C}{p_C^C}\right)\Big|_{t=0} = 1-\Lambda > 0. \tag{28}$$

In market D, $\ln(p_D^D/p_C^D) = \ln p_D - \ln\tau_{CD} - \ln p_C$, so

$$\frac{\partial}{\partial t}\ln\left(\frac{p_D^D}{p_C^D}\right)\Big|_{t=0} = -\Lambda < 0. \tag{29}$$

Both signs follow from $\Lambda \in (0,1)$.

Step 4: Share responses and parts (a) and (b)

From (27), the ratio of quantities in market j is $q_{Dj}/q_{Cj} = (\theta_{Dj}/\theta_{Cj})(p_D^j/p_C^j)^{-\sigma}$. Writing $\tilde{s}_{Dj} = r_j/(1+r_j)$ with $r_j \equiv q_{Dj}/q_{Cj}$ and using $d\tilde{s}_{Dj} = \tilde{s}_{Dj}(1-\tilde{s}_{Dj})\,d\ln r_j$,

$$\frac{\partial\tilde{s}_{Dj}}{\partial t} = -\sigma\,\tilde{s}_{Dj}(1-\tilde{s}_{Dj})\,\frac{\partial}{\partial t}\ln\left(\frac{p_D^j}{p_C^j}\right). \tag{30}$$

The coefficient is $-\sigma$ rather than $1-\sigma$ because $\tilde{s}_{Dj}$ is a quantity share; if shares are instead measured in expenditure, $-\sigma$ is replaced by $1-\sigma$ throughout. Since $\sigma > 1$, both are negative and none of what follows is affected.[17] Combining (30) with (28) and (29) gives $\partial\tilde{s}_{DC}/\partial t < 0$ and $\partial\tilde{s}_{DD}/\partial t > 0$, which proves parts (a) and (b).

Step 5: The global share and condition (17)

Because X_C and X_D are fixed, the destination weights μ_j in (16) are fixed and the destination-mix term vanishes. The within-market component is

$$\sum_j \mu_j \frac{\partial\tilde{s}_{Dj}}{\partial t} = -\sigma\Big[\underbrace{\mu_C\,\tilde{s}_{DC}(1-\tilde{s}_{DC})}_{A}(1-\Lambda) - \underbrace{\mu_D\,\tilde{s}_{DD}(1-\tilde{s}_{DD})}_{B}\Lambda\Big].$$

[17] Condition (17) is invariant to this choice, since the elasticity factors out of both terms.

With $-\sigma < 0$, this is positive if and only if the bracket is negative, that is if and only if $B\Lambda > A(1-\Lambda)$. Rearranging, $\Lambda(A+B) > A$, or $\Lambda > A/(A+B)$, which is condition (17).

Step 6: The link to the decomposition

Let $\varsigma_i = q_i/q$ denote global quantity shares, $E_i = e_i q_i$ the emissions of origin i, and $E = \sum_i E_i$. In a single sector with two origins, $\varsigma_C + \varsigma_D = 1$ implies $d\varsigma_C = -d\varsigma_D$, and the sourcing term of equation (10) is

$$\sum_i \frac{E_i}{E}\, d\ln \varsigma_i = \sum_i \frac{e_i q\, \varsigma_i}{E} \cdot \frac{d\varsigma_i}{\varsigma_i} = \frac{q}{E}\sum_i e_i\, d\varsigma_i = \frac{q\,(e_D - e_C)}{E}\, d\varsigma_D,$$

which is equation (18). The factor $q(e_D - e_C)/E$ is strictly positive under $e_C < e_D$, so the sourcing term is positive exactly when $d\varsigma_D > 0$. This identity is exact and does not depend on the levels of Y_C and Y_D. In particular, it does not require taking a stand on how gross output is split between final and intermediate uses. Combining it with Step 5 completes the proof of the proposition. ■

Proof of Corollary 1

Let D impose a tariff t_R on shipments from C, so that $p_C^D = \tau_{CD}\, p_C\,(1+t_R)$, and write $\rho = \partial t_R/\partial t \geq 0$. Since $\omega_{CD} = 0$, country D does not use C's output as an input and $d\ln p_D = 0$ still holds, so Step 2 is unchanged. Market C is unaffected by t_R and (28) continues to hold. In market D,

$$\frac{\partial}{\partial t}\ln\left(\frac{p_D^D}{p_C^D}\right)\Big|_{t=0} = -(\Lambda+\rho) < -\Lambda < 0,$$

so part (b) is reinforced and part (a) is unchanged. Repeating Step 5, the within-market component is positive if and only if $B(\Lambda+\rho) > A(1-\Lambda)$, equivalently

$$\Lambda + \frac{B}{A+B}\,\rho \;>\; \frac{A}{A+B}.$$

Under reciprocal retaliation $\rho = 1$ this reduces to $\Lambda > (A-B)/(A+B)$, which holds for any $\Lambda > 0$ whenever $B \geq A$. ■

Extension 1: the intermediate margin

Let f_{ij} denote the total flow of origin i's variety into market j, comprising final demand q_{ij} with elasticity σ and intermediate demand m_{ij} with elasticity η, and let $\tilde{s}_{Dj}$ be defined over f_{ij}. Because the tariff applies

to final and intermediate shipments alike, both margins face the same delivered relative price and therefore the same right-hand sides (28) and (29). Letting ϕ_j be the share of final demand in total flows into market j, and evaluating at a point where the dirty origin's share is common to the two margins,

$$\frac{\partial \tilde{s}_{Dj}}{\partial t} = -\big[\phi_j \sigma + (1-\phi_j)\eta\big]\, \tilde{s}_{Dj}(1-\tilde{s}_{Dj})\, \frac{\partial}{\partial t} \ln\left(\frac{p_D^j}{p_C^j}\right).$$

Since $\sigma, \eta > 1$, the bracket is positive and parts (a) and (b) go through unchanged. Condition (17) is unaffected whenever the bracket is common across destinations, and otherwise A and B are simply reweighted by their respective brackets.

Extension 2: relaxing $\omega_{CD} = 0$

Allow $\omega_{CD} > 0$, so that D uses C's output as an input. The two pass-through equations become

$$d\ln p_C = \alpha\big[\bar{\omega}_{CC}\, d\ln p_C + \bar{\omega}_{DC}(d\ln p_D + dt)\big], \qquad d\ln p_D = \alpha\big[\bar{\omega}_{CD}\, d\ln p_C + \bar{\omega}_{DD}\, d\ln p_D\big].$$

The second gives $d\ln p_D = \delta\, d\ln p_C$ with

$$\delta \equiv \frac{\alpha\bar{\omega}_{CD}}{1-\alpha\bar{\omega}_{DD}} \in (0,1),$$

and substituting into the first yields

$$\left.\frac{\partial \ln p_C}{\partial t}\right|_{t=0} = \Lambda^* \equiv \frac{\alpha\bar{\omega}_{DC}}{1-\alpha\bar{\omega}_{CC}-\alpha\bar{\omega}_{DC}\,\delta} > 0.$$

The relative-price responses become $1-(1-\delta)\Lambda^*$ in market C and $-(1-\delta)\Lambda^*$ in market D. Hence every statement above holds with Λ replaced by

$$\tilde{\Lambda} \equiv (1-\delta)\,\Lambda^*.$$

It remains to check $\tilde{\Lambda} \in (0,1)$. Positivity follows from $\delta < 1$ and $\Lambda^* > 0$, the latter because $\alpha\bar{\omega}_{CC} + \alpha\bar{\omega}_{DC}\delta < \alpha(\bar{\omega}_{CC} + \bar{\omega}_{DC}) = \alpha < 1$. For the upper bound, $\tilde{\Lambda} < 1$ is equivalent to $\alpha\bar{\omega}_{DC}(1-\delta) < 1 - \alpha\bar{\omega}_{CC} - \alpha\bar{\omega}_{DC}\delta$, which simplifies to $\alpha(\bar{\omega}_{DC} + \bar{\omega}_{CC}) < 1$, that is $\alpha < 1$. Parts (a) and (b) therefore survive for any $\omega_{CD} \geq 0$, and condition (17) holds with Λ replaced by $\tilde{\Lambda}$. Since $\tilde{\Lambda} < \Lambda$, the feedback through D's costs attenuates both the green protection and the brown sourcing force, the attenuation being governed by δ, which is of the order of D's reliance on inputs from C.

F Solving algorithm

In this section, we detail how we reduce the model in changes presented above to a $N \times S$ system that we use to back out counterfactual results.

Equations (19) and (20) imply that :

$$\widetilde{q}_{is'j} = \widetilde{I}_j \widetilde{q}_{is'j}{}^{\circ} \tag{31}$$

where $\widetilde{q}_{is'j}{}^{\circ} = [\widetilde{p}_{is'} \frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{-\sigma_{s'}} \widetilde{P}^{c\ (\sigma_{s'}-1)}_{s'j}$ only depends on the change of prices and the baseline. It is useful to note that this inverse is linear in any change of set of prices : $\widetilde{q}_{is'j}{}^{\circ}(\alpha\widetilde{p}) = \widetilde{q}_{is'j}{}^{\circ}(\widetilde{p})/\alpha$. This is the most general expression of the change of quantities traded so that the condition of consumer spending is respected by construction because : $\sum_{i,s'} \widetilde{p}_{is'}(1+t_{is'j})\widetilde{q}_{is'j}{}^{\circ}(p^B_{is'j}q^B_{is'j}) = I^B_j$. The form of this expression represents that if the income of the consumer increases (or decreases), she will proportionally increase his consumption from every economy/sector. $\widetilde{q}_{is'j}{}^{\circ}$ contains all the information of the reorganisation of his consumption if his income did not change in the counterfactual world.

Similarly, for intermediates, equation (21) together with $\widetilde{E}_{s'js} = \widetilde{E}_{js}$ imply that :

$$\widetilde{m}_{is'js} = \widetilde{E}_{js} \widetilde{m}_{is'js}{}^{\circ} \tag{32}$$

where $\widetilde{m}_{is'js}{}^{\circ} = [\widetilde{p}_{is'} \frac{(1+t_{is'j})}{(1+t^B_{is'j})}]^{-\eta_{s's}} \widetilde{P}^{p\ (\eta_{s's}-1)}_{s'js}$ has the same properties as $\widetilde{q}_{is'j}{}^{\circ}$. The construction makes sure that the producer spending is respected.

Having the consumer and producer spending respected by construction, we need to compute the consumer and producer revenue. We use equations (26) with $\widetilde{E}_{js} = \widetilde{Y}_{js}$ to compute the wages change under a change of spending of the producer:

$$\widetilde{w}_j = \frac{\sum_s \widetilde{E}_{js} L^B_{js}}{L_j} \tag{33}$$

We have then made sure that the solution respects the labor market clearing condition and the constitutive equation of production $L_{js} = \gamma_{j,Ls} \frac{Y_{js}}{w_j}$, and we can write the consumer revenue and producer spending from the consumer and producer clearing equations :

$$\widetilde{I}_j I_j^B = \sum_s \widetilde{E}_{js} L_{js}^B w_j^B + \sum_{i,s'} \widetilde{I}_j \widetilde{p}_{is'} t_{is'j} \widetilde{q}_{is'j}{}^{\circ}(p_{is'j}^B q_{is'j}^B) + \sum_{i,s',s} \widetilde{E}_{js} \widetilde{p}_{is'} t_{is'j} \widetilde{m}_{is'js}{}^{\circ}(p_{is'j}^B m_{is'js}^B) + D_j^B \tag{34}$$

$$\widetilde{E}_{is'} E_{is'}^B = \sum_j \widetilde{I}_j \widetilde{p}_{is'} \widetilde{q}_{is'j}{}^{\circ}(p_{is'j}^B q_{is'j}^B) + \sum_{j,s} \widetilde{E}_{js} \widetilde{p}_{is'} \widetilde{m}_{is'js}{}^{\circ}(p_{is'j}^B m_{is'js}^B) \tag{35}$$

We then use (34) in (35):

$$\widetilde{E}_{is'} E_{is'}^B = \widetilde{p}_{is'} \left(\sum_{j,s} \widetilde{E}_{js} \left[\widetilde{q}_{is'j}{}^{\circ}(p_{is'j}^B q_{is'j}^B) \frac{L_{js}^B w_j^B + \sum_{i,s'} \widetilde{m}_{is'js}{}^{\circ} \widetilde{p}_{is'} t_{is'j} (p_{is'j}^B q_{is'j}^B)}{I_j^B - \sum_{i,s'} \widetilde{q}_{is'j}{}^{\circ} \widetilde{p}_{is'} t_{is'j} (p_{is'j}^B q_{is'j}^B)} + \widetilde{m}_{is'js}{}^{\circ}(p_{is'j}^B m_{is'js}^B) \right] + \sum_j \frac{D_j \widetilde{q}_{is'j}{}^{\circ}(p_{is'j}^B q_{is'j}^B)}{I_j^B - \sum_{i,s'} \widetilde{p}_{is'} t_{is'j} \widetilde{q}_{is'j}{}^{\circ}(p_{is'j}^B q_{is'j}^B)} \right) \tag{36}$$

$$\widetilde{p}_{js} = \left(\sum_{s'} \widetilde{E}_{js'} \frac{L_{js'}^B}{L_j} \right)^{\gamma_{j,Ls}} \prod_{s'} \widetilde{P}_{s'js}^{\;p \quad \gamma_{s'js}} \tag{37}$$

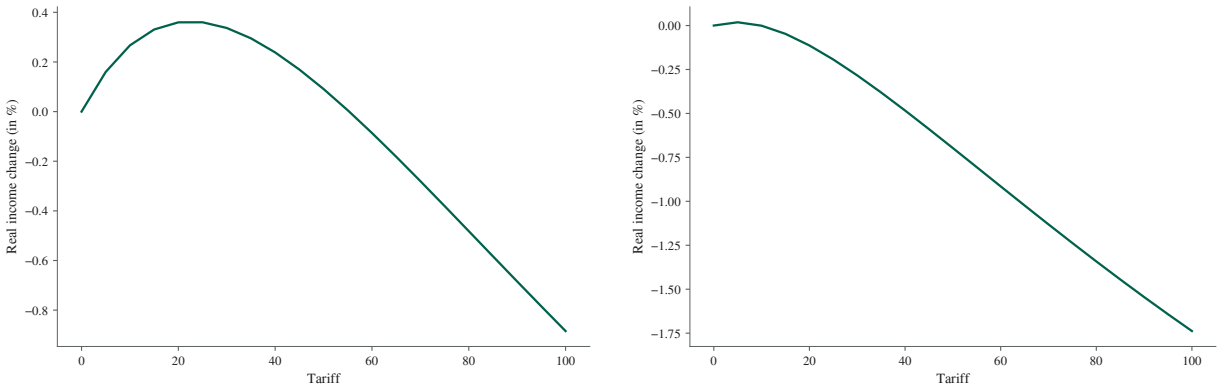
with the last equation expressing the cost of production from the solution of the cost minimization of the production costs of the producer (23). We have thus reduced the equations to a system of two non-linear equations (36) and (37) of the two fundamental hat quantities $(\widetilde{E}, \widetilde{p})$. Since we have explicit expressions of the variables on the right hand side, we can solve numerically this system with a nested fixed point routine.

The solution space of this system of equations is of dimension 1, any linear transformation $\alpha(\widetilde{E}_{\text{sol}}, \widetilde{p}_{\text{sol}})$ of a solution of the system is also solution. We need to add one numeraire constraint to make the solution unique. Our benchmark results use the global average wage as the numeraire.

F.1 The optimal and revenue-maximizing tariff

Figure 11(a) traces U.S. real income as a uniform tariff on all partners rises: it increases up to a peak near 25% and declines thereafter, while under symmetric retaliation (panel b) the optimal tariff is only 5%. This welfare-maximizing rate is above the 19% uniform optimum in Ignatenko et al. (2025). The gap follows from two differences noted earlier. Our estimated substitution elasticities (mean $\sigma \approx 3.61$) imply a lower trade elasticity than their $\varepsilon = 4$, and we have no markup or selection margin, whereas firm selection makes trade more tariff-sensitive in their model. Both make a marginal tariff less distortionary and push the optimal rate up. A low U.S. import share reinforces this: under a uniform schedule imports can be displaced only

Figure 11: Real Income Change in the U.S. under Unilateral and Retaliatory Tariffs



(a) No retaliation

(b) Retaliatory tariffs

Notes: Change in U.S. real income, in percent relative to the January 2025 baseline, as a uniform ad valorem tariff applied to all partners and all sectors rises from 0 to 100%. Panel (a) shows the unilateral case, in which no partner retaliates; real income peaks near a 25% rate. Panel (b) shows symmetric retaliation, in which every country imposes the same rate on U.S. imports; real income rises slightly up to a rate of 5% and decreases thereafter. Tariff revenue is rebated lump-sum throughout.

by domestic production, so the base erodes slowly.

Figure 12 plots revenue and real income against a common uniform rate. The revenue-maximizing tariff, where the elasticity of dutiable import value with respect to the tariff reaches -1, lies far above the welfare-maximizing tariff, where the marginal terms-of-trade gain equals the marginal deadweight loss. The two need not coincide, and here revenue peaks only at a uniform rate of 85%, against 25% for real income: because a uniform schedule displaces imports only toward domestic production, the base contracts slowly and revenue turns down only at high rates.

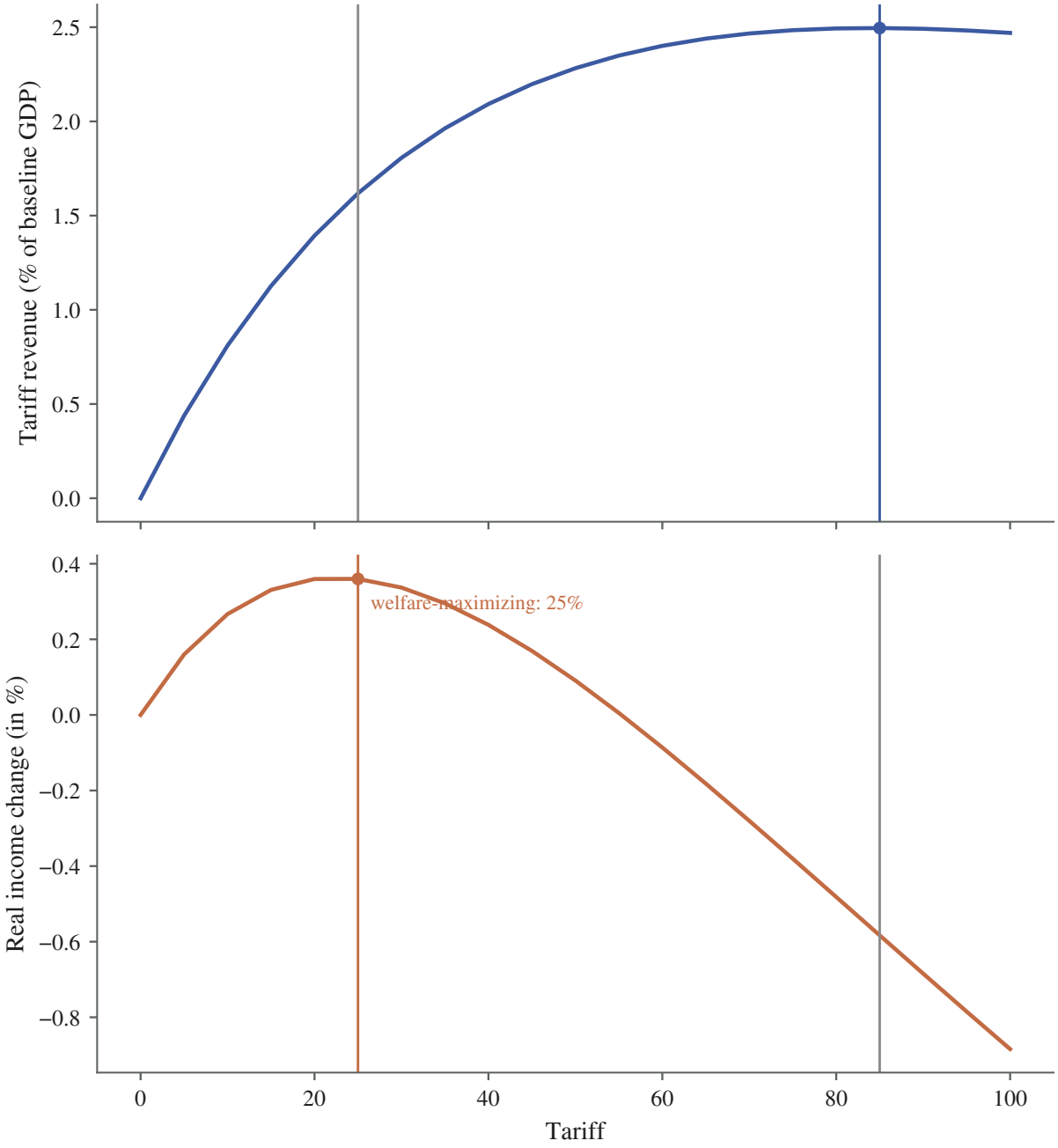
F.2 Lost tariff revenue exercise

Our baseline follows standard practice in the quantitative trade literature and rebates tariff revenue to the representative household as a lump-sum transfer, so that R_j enters income in equation 4. We compare the results from the standard model to the case where tariff revenue is lost.

We implement this as an ex-post correction to household income rather than by re-solving the model with a modified income equation. This is deliberate and, under our assumptions, exact. The equilibrium conditions that determine relative prices, expenditure shares and the allocation of production in equations 3, 7, 8 and 9 depend on income only through the scalar I_j, which enters demand multiplicatively.

Moreover, the upper tier of preferences in equation 1 is Cobb-Douglas, so the expenditure shares $\beta_{s'j}$ are independent of income and there is no income effect on the composition of demand. Together these imply

Figure 12: Tariff revenue and real income change - Uniform tariffs



Notes: U.S. tariff revenue as a share of baseline GDP (upper panel) and the change in U.S. real income in percent (lower panel), plotted against a common uniform ad valorem tariff applied to all partners and all sectors. The vertical line marks the welfare-maximizing rate.

that the trade distortion generated by the tariff is invariant to how the revenue is subsequently disposed of within the destination country: what changes when revenue is dissipated is the level of resources available to the U.S. household, not the pattern of trade. This is what makes the correction below exact rather than

a first-order approximation[18].

Concretely, we proceed in three steps. First, we solve the counterfactual equilibrium exactly as in the baseline, obtaining the counterfactual final consumption quantities $q^{cf}_{is'j}$ and intermediate flows $m^{cf}_{is'js}$ at counterfactual prices. Second, we compute the tariff revenue accruing to the U.S. government from the newly imposed tariffs,

$$\Delta R_{US} \;=\; \sum_{i,s'} t_{is'US}\, p_{is'US}\, q^{cf}_{is'US} \;+\; \sum_{i,s',s} t_{is'US}\, p_{is'US}\, m^{cf}_{is'USs}, \tag{38}$$

where $t_{is'US}$ denotes the counterfactual tariff schedule. We measure revenue from the counterfactual schedule only, since revenue from the January 2025 applied tariffs is already embedded in the baseline level of I_{US} against which changes are measured, including it would double-count. Third, we subtract this revenue from U.S. income and rescale welfare,

$$\hat{I}^{norev}_{US} \;=\; \hat{I}_{US} \;-\; \frac{\Delta R_{US}}{I^{B}_{US}}, \qquad \hat{U}^{norev}_{US} \;=\; \hat{U}_{US}\, \frac{\hat{I}^{norev}_{US}}{\hat{I}_{US}}, \tag{39}$$

where I^{B}_{US} is baseline U.S. income and hats denote counterfactual-to-baseline ratios. The proportional rescaling is exact: preferences (1)–(2) are homothetic, so indirect utility is $U_j = I_j/P_j$ with P_j the Cobb-Douglas-CES price index, and P_{US} is unchanged by construction. Welfare in all other countries is identical to the baseline counterfactual, since only the intra-U.S. disposition of revenue differs.

G Elasticity Estimation

For details on the elasticity estimation, see Le Moigne et al. (2026). Table 3 shows the elasticities used in the paper. CP stands for the estimation method in Caliendo and Parro (2015), FG for the estimation method in Fontagné et al. (2022), and Resc. for the elasticities taken directly from Fontagné et al. (2022) and aggregated to our sectoral level.

[18]Re-solving with the revenue term deleted from equation 4 would additionally violate the income–expenditure consistency on which the fixed-point iteration relies, and the solver does not converge. The ex-post correction avoids this while delivering the object of interest.

Table 3: Elasticities

Sector	Code	CP	Resc.	FGO	Sector	Code	CP	Resc.	FGO
Agriculture	01T02	4.95	2.89	8.27	Water supply	36T39	1.91	6.35	5.05
Fishing	03	2.58	3.28	2.47	Construction	41T43	3.49	2.67	4.95
Mining, energy	05T06	1.78	6.72	5.08	Wholesale, retail	45T47	3.49	2.67	4.95
Mining, non-energy	07T08	3.29	8.84	4.58	Land transport	49	3.49	2.67	4.95
Food products	10T12	4.05	2.93	6.31	Water transport	50	3.49	2.67	4.95
Textiles	13T15	5.17	2.86	5.75	Air transport	51	3.49	2.67	4.95
Wood	16	5.02	4.60	5.69	Warehousing	52	3.49	2.67	4.95
Paper	17T18	4.54	4.78	6.86	Post	53	3.49	2.67	4.95
Coke, petroleum	19	3.61	4.82	5.07	Tourism	55T56	3.49	2.67	4.95
Chemicals	20	3.77	4.87	6.72	Media	58T60	4.03	4.72	5.15
Pharmaceuticals	21	4.10	7.81	5.33	Telecom	61	3.49	2.67	4.95
Plastics	22	4.17	3.59	6.19	IT	62T63	3.49	2.67	4.95
Non-metallic minerals	23	3.68	3.25	5.36	Finance, insurance	64T66	3.49	2.67	4.95
Basic metals	24	5.86	5.87	7.19	Real estate	68	3.49	2.67	4.95
Fabricated metals	25	4.87	2.35	5.89	R&D	69T75	3.76	4.96	2.38
Electronics	26	2.19	2.78	5.63	Administration	77T82	2.46	4.84	2.62
Electrical equipment	27	3.24	2.41	6.00	Public sector	84	3.49	2.67	4.95
Machinery	28	2.91	4.49	6.02	Education	85	2.80	4.79	2.78
Transport equipment	29T30	3.49	5.01	4.95	Health	86T88	3.49	2.67	4.95
Manufacturing nec	31T33	4.90	3.94	6.56	Entertainment	90T93	2.29	4.84	3.22
Energy	35	3.78	9.32	1.71	Other service	94T98	3.49	2.67	4.95

Notes: Elasticities of substitution across origins, by OECD ICIO sector. The model imposes $\sigma_{s'} = \eta_{s'}$, so a single elasticity governs both final and intermediate substitution. "CP" applies the estimation method of Caliendo and Parro (2015) to our sample; "FGO" applies the method of Fontagné et al. (2022) to the same sample, supplemented with their tariff data; "Resc." takes the product-level elasticities of Fontagné et al. (2022) directly and aggregates them to our sectoral definitions. Estimates that are negative, statistically insignificant, or not identified for a lack of tariff variation—predominantly in service sectors—are replaced by the sample mean of the corresponding column. CP is the baseline throughout; FGO and Resc. are used in the robustness exercises.

H Territorial vs. footprint emissions construction

Territorial emissions. At each country-sector (i, s), physical emission intensity e_{is} (baseline tCO_2 per unit of real output) is held fixed. Territorial (production-based) emissions of country i are

$$E_i = \sum_s e_{is}\, Y_{is}, \tag{40}$$

where Y_{is} is real (price-deflated) gross output — exactly the quantity already reported as `sol.co2_prod`, summed over sectors.

Footprint emissions. Consumption-based (footprint) emissions attribute production-side emissions to whichever country ultimately consumes the good, following it through every tier of the supply chain. Let A be the technical-coefficient matrix rebuilt from real (price-deflated) bilateral intermediate flows (counterfactual A for counterfactual footprints; baseline A for the baseline reference, where real and nominal coincide since baseline $\hat{p} \equiv 1$), and $L = (I - A)^{-1}$ the associated Leontief inverse. With f_{isj} the real bilateral final-consumption flow from origin (i, s) to destination j, footprint emissions of country j are

$$F_j = \sum_{i,s} \left[e'L\right]_{is} f_{isj}, \tag{41}$$

i.e. the embodied-intensity vector $e'L$ (emissions per unit of final demand for each good, direct plus every indirect tier) applied to the bundle of goods j actually consumes, regardless of where they were produced.

Because a Leontief demand-driven system only reallocates emissions across countries and cannot create or destroy them, footprint and territorial emissions reconcile exactly at the global total, $\sum_j F_j = \sum_i E_i$ — confirmed to hold with zero discrepancy at baseline and to solver-tolerance noise ($\sim 10^{-5}$ Mt) in every counterfactual.

The gap. For each country, comparing its own footprint change to its own territorial change isolates who is "exporting" vs. "importing" embodied emissions:

$$\text{gap}_j = \Delta F_j - \Delta E_j, \qquad \Delta X_j = X_j^{cf} - X_j^{bl}. \tag{42}$$

I Country-level footprint decomposition

The green-sourcing term in the main text (Section 3.3) is a global, production-side object: it decomposes the change in the aggregate origin-substitution term using each origin country's global market share. This appendix reports a complementary, country-level, consumption-side decomposition: for each destination country j, how much of the change in its consumption-based (footprint) emissions is driven by (i) how much it consumes, (ii) what it consumes, and (iii) where and how cleanly that consumption is sourced.

I.1 Building blocks

Let i, j index countries and s index sectors. Let F_{isj} denote the real (price-deflated) bilateral final-consumption flow from origin i, sector s, to destination j, and let ℓ_{is} be the total embodied emissions intensity of country-sector (i, s),

$$\ell_{is} = \left[e'(I - A)^{-1}\right]_{is}, \tag{43}$$

where e is the vector of direct (physical) emissions intensities and A is the technical-coefficient matrix rebuilt from real intermediate flows (the same construction used for the territorial-vs-footprint reconciliation in the main text: counterfactual A for counterfactual ℓ, baseline A for baseline ℓ).

From F_{isj} and ℓ_{is} we define, for each destination j:

$$f_j = \sum_{i,s} F_{isj} \qquad \text{(total real consumption absorbed by } j\text{)} \tag{44}$$

$$\phi_{sj} = \frac{\sum_i F_{isj}}{f_j} \qquad \text{(sector-}s\text{ share of } j\text{'s consumption basket)} \tag{45}$$

$$\mu_{isj} = \frac{F_{isj}}{\sum_{i'} F_{i'sj}} \qquad \text{(origin-}i\text{ share of } j\text{'s sector-}s\text{ purchases)} \tag{46}$$

$$\varepsilon_{sj} = \sum_i \mu_{isj}\, \ell_{is} \qquad \text{(embodied intensity of } j\text{'s sector-}s\text{ purchases)} \tag{47}$$

$$\bar{e}_j = \sum_s \phi_{sj}\, \varepsilon_{sj} \qquad \text{(average embodied intensity of } j\text{'s basket)} \tag{48}$$

Country j's footprint is then $F_j = f_j\, \bar{e}_j$, which reproduces the footprint measure used in the main text's territorial-vs-footprint figures exactly.

I.2 Exact decomposition

We want to split $\Delta F_j = F_j^{cf} - F_j^{bl}$ into a *scale* effect (more/less total consumption), a *composition* effect (a dirtier/cleaner sectoral mix), and a *sourcing* effect (dirtier/cleaner origins and embodied technology within

each sector). Because F_j is a nested product of several terms, a decomposition weighted by baseline values only (as used for the global term in the main text) is not exact here — it left a residual exceeding 2% of ΔF_j for the top-20 emitters in all 13 scenarios (mean residual up to 12% of ΔF_j, occasionally exceeding 100% for countries with small ΔF_j). We therefore use the *exact* two-factor midpoint identity

$$\Delta(ab) = \tfrac{1}{2}(a^{bl} + a^{cf})\,\Delta b + \tfrac{1}{2}(b^{bl} + b^{cf})\,\Delta a, \tag{49}$$

which holds with *zero* residual for any two scalars a, b (it is the average of the two orderings "a-then-b" and "b-then-a" in a discrete product-rule split, i.e. the 2-player Shapley value of the change in ab). Applying (49) at each nesting level of $F_j = f_j \sum_s \phi_{sj}\varepsilon_{sj}$ gives, with $^{-}$ denoting the midpoint of baseline and counterfactual,

$$\text{scale}_j = \bar{e}_j^{\,mid}\,\Delta f_j, \tag{50}$$

$$\text{composition}_j = f_j^{\,mid} \sum_s \varepsilon_{sj}^{\,mid}\,\Delta\phi_{sj}, \tag{51}$$

$$\text{sourcing}_j = f_j^{\,mid} \sum_s \phi_{sj}^{\,mid}\,\Delta\varepsilon_{sj}, \tag{52}$$

so that $\text{scale}_j + \text{composition}_j + \text{sourcing}_j = \Delta F_j$ exactly, for every country and every scenario (verified to machine precision, $\leq 10^{-11}$ Mt).

I.3 Sourcing sub-split: origin mix vs. embodied technology

The sourcing term further splits into how much comes from countries reshuffling *which* origins they buy from at fixed technology (μ) versus how *dirty those origins already were* changing (ℓ, via each origin's own Leontief inverse). Applying (49) once more to $\varepsilon_{sj} = \sum_i \mu_{isj}\,\ell_{is}$:

$$\text{direct}_{sj} = \sum_i \ell_{is}^{\,mid}\,\Delta\mu_{isj} \qquad \text{(origin-mix substitution)} \tag{53}$$

$$\text{upstream}_{sj} = \sum_i \mu_{isj}^{\,mid}\,\Delta\ell_{is} \qquad \text{(embodied-intensity / supply-chain propagation)} \tag{54}$$

aggregated to the country level with the same weights as sourcing_j:

$$\text{direct}_j = f_j^{\,mid} \sum_s \phi_{sj}^{\,mid}\,\text{direct}_{sj}, \qquad \text{upstream}_j = f_j^{\,mid} \sum_s \phi_{sj}^{\,mid}\,\text{upstream}_{sj}, \tag{55}$$

so that $\text{direct}_j + \text{upstream}_j = \text{sourcing}_j$ exactly (verified). A country's sourcing term moving because it switched suppliers within a sector shows up in direct_j; it moving because its existing suppliers' own upstream

footprints changed (e.g. a supplier's own imports got cleaner) shows up in upstream_j.

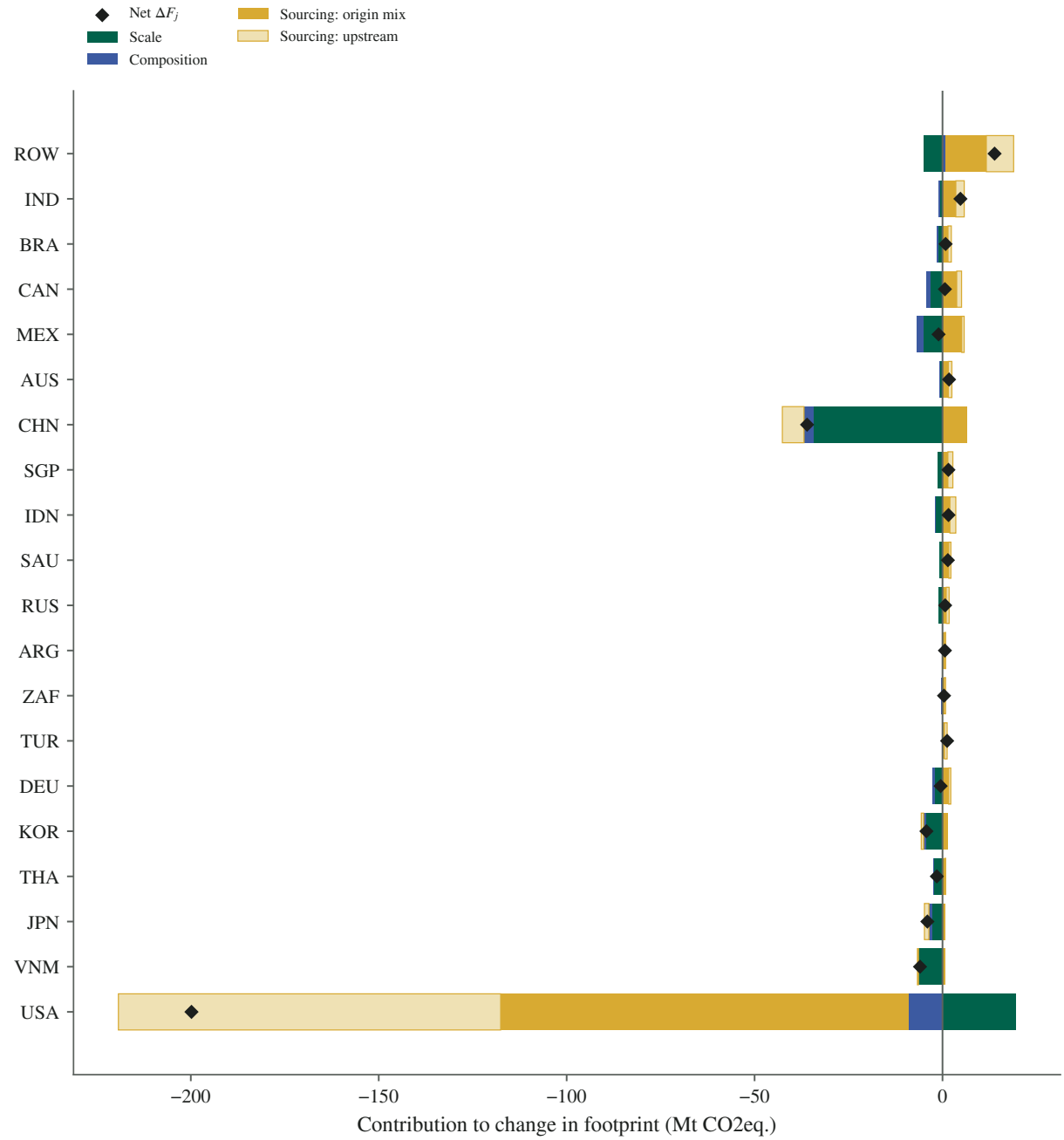
Figure 13 shows the decomposition under Liberation Day for the twenty largest baseline emitters and the rest of the world. Each bar splits the change in the footprint into scale_j, composition_j, and the two parts of the sourcing term, direct_j (origin mix) and upstream_j. A diamond marks the net ΔF_j, which equals the sum of the four parts by construction.

Two results stand out. First, the large fall in the U.S. footprint (−199.8 Mt) is a sourcing effect, not a scale effect. U.S. real final consumption rises by 0.33%, so scale_j is positive (+19.4 Mt) and composition_j is small (−9.0 Mt). The sourcing term (−210.3 Mt) splits almost evenly into origin mix (−108.6 Mt) and upstream (−101.7 Mt). U.S. consumers thus buy from cleaner origins, and the suppliers they already buy from use cleaner inputs. Second, the footprint rises in 51 economies, by 47.0 Mt in total. Sourcing raises their footprints by 67.1 Mt (39.4 Mt origin mix and 27.6 Mt upstream), while scale lowers them by 19.6 Mt and composition by 0.5 Mt. In these economies, the rise in the footprint therefore comes from dirtier sourcing, both in their own purchases and further up their supply chains.

Figure 14 reports sourcing_j for the same economies in all 13 scenarios. The pattern of Liberation Day holds throughout. The U.S. sourcing term is negative in every scenario, between −339.2 Mt (US2060Retal) and −111.6 Mt (US10). Of the 19 other economies, 16 have a positive sourcing term in every scenario. The exceptions are China, Germany, and Japan, whose terms are negative in at least one scenario. In the campaign scenarios, retaliation makes the dirtier sourcing of other economies larger. Under US2060, for example, the rest of the world's sourcing term rises from 22.1 to 35.5 Mt and China's from 7.1 to 32.5 Mt.

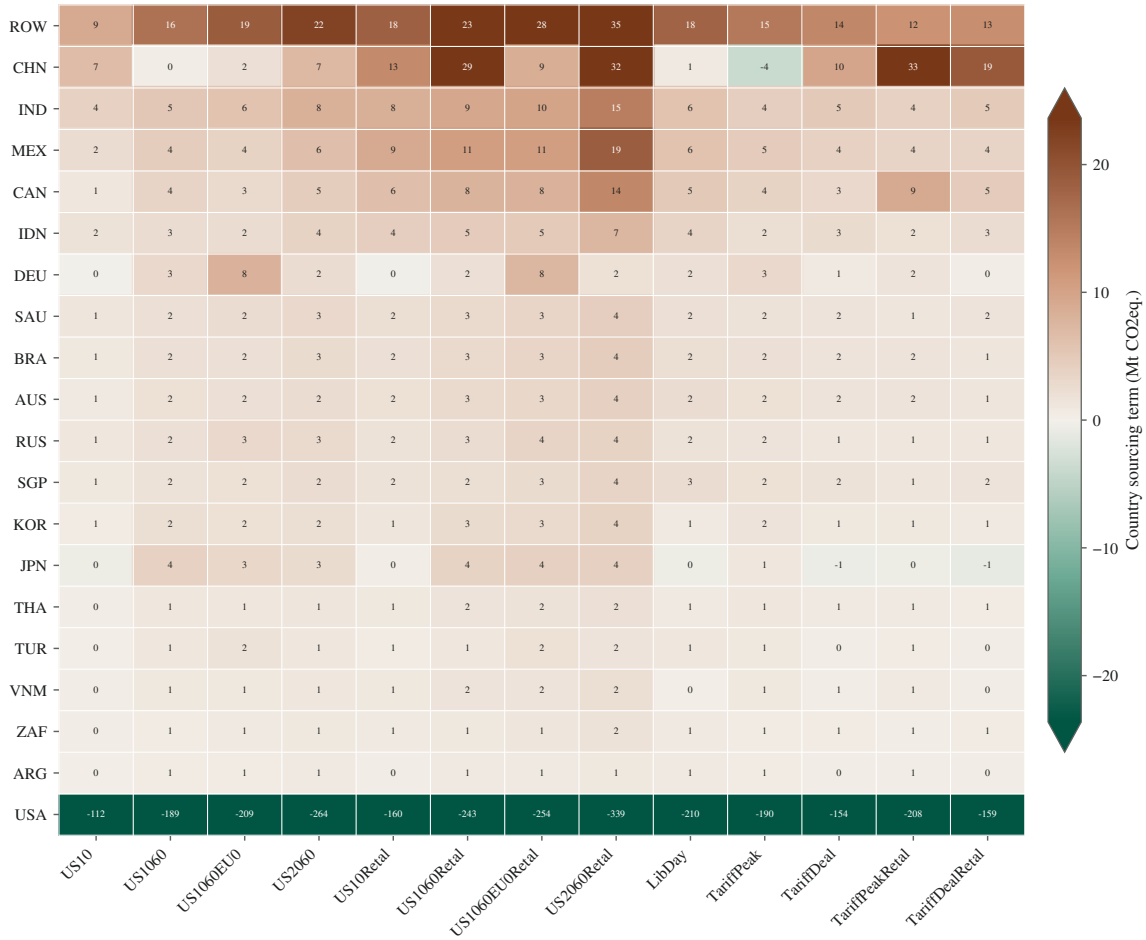
Caveat. The country-level sourcing_j terms summed across countries are *not* the same object as the global green-sourcing term (Figure 5) reported in the main text: one is a consumption-side, destination-country sum; the other is a production-side, origin-country share decomposition. The two are positively correlated across scenarios but should not be cited as reproducing one another.

Figure 13: Country-level footprint decomposition Liberation Day



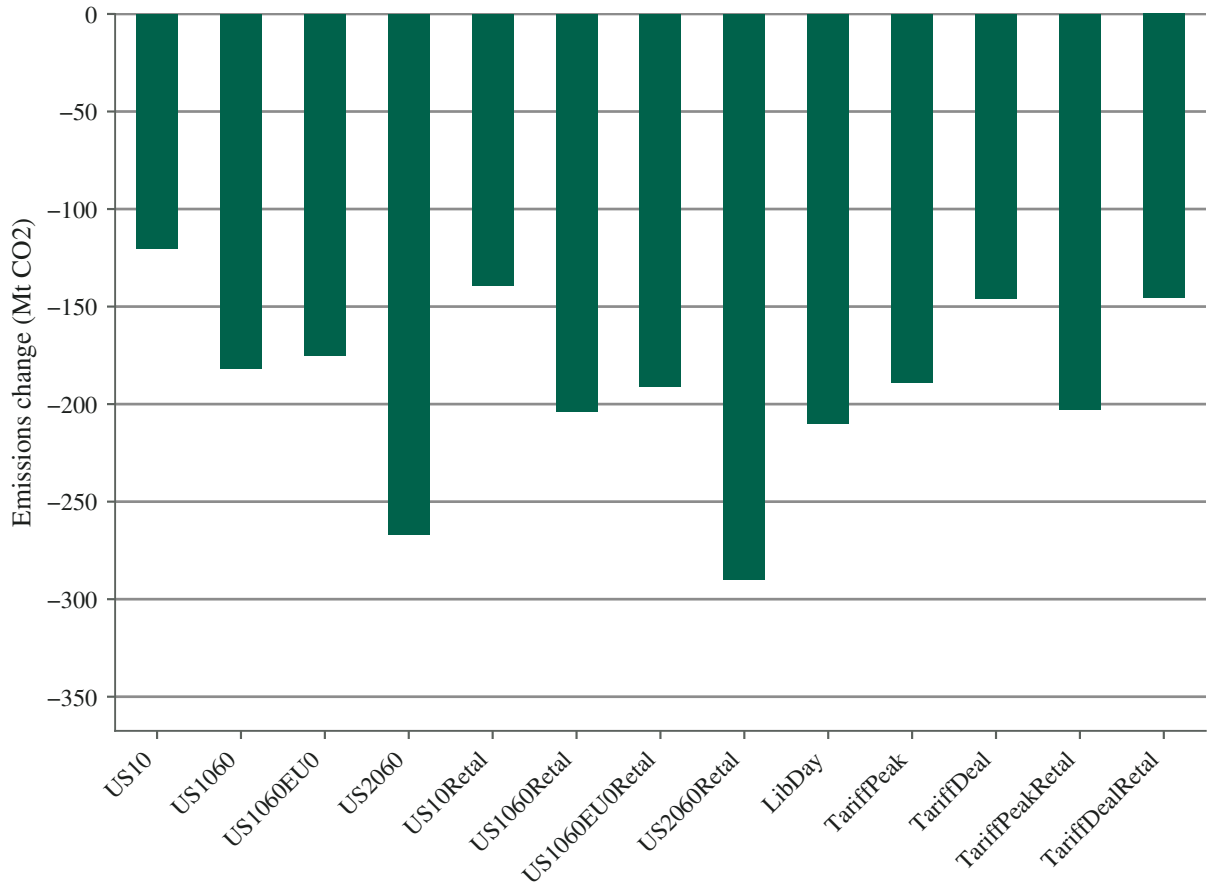
Notes: Decomposition of the change in each country's consumption-based (footprint) emissions under the Liberation Day schedule, in Mt CO_2eq. The scale term measures how much total real consumption changes, the composition term how the sectoral mix of that consumption changes, and the sourcing term how the origin mix and the embodied intensity of suppliers change. The sourcing term is shown in two parts: the origin mix (full colour), which holds suppliers' embodied intensity fixed, and the upstream part (pale, outlined), which holds the country's own origin shares fixed. Positive and negative parts are stacked separately from zero. The split uses the exact two-factor midpoint (Shapley) identity, so the four components sum to the net change, marked by the diamond, with zero residual. The plotted economies are the twenty largest baseline territorial emitters plus the ICIO rest-of-world region. The country-level sourcing terms summed across destinations are a consumption-side object and are not the same as the production-side green-sourcing term of Figure 5; the two are positively correlated across scenarios but should not be read as reproducing one another.

Figure 14: Footprint sourcing term by country, all scenarios



Notes: Sourcing term of the country-level footprint decomposition, $\text{sourcing}_j = \text{direct}_j + \text{upstream}_j$, in Mt CO_2eq., for each scenario. Positive values (brown) mean that dirtier sourcing raises the country's footprint, negative values (green) that cleaner sourcing lowers it. Each cell shows the value rounded to whole Mt. The colour scale is set by the economies other than the United States, so the U.S. cells, which are an order of magnitude larger, are shown at the darkest shade. The plotted economies are the twenty largest baseline territorial emitters plus the ICIO rest-of-world region, ordered by their mean across scenarios.

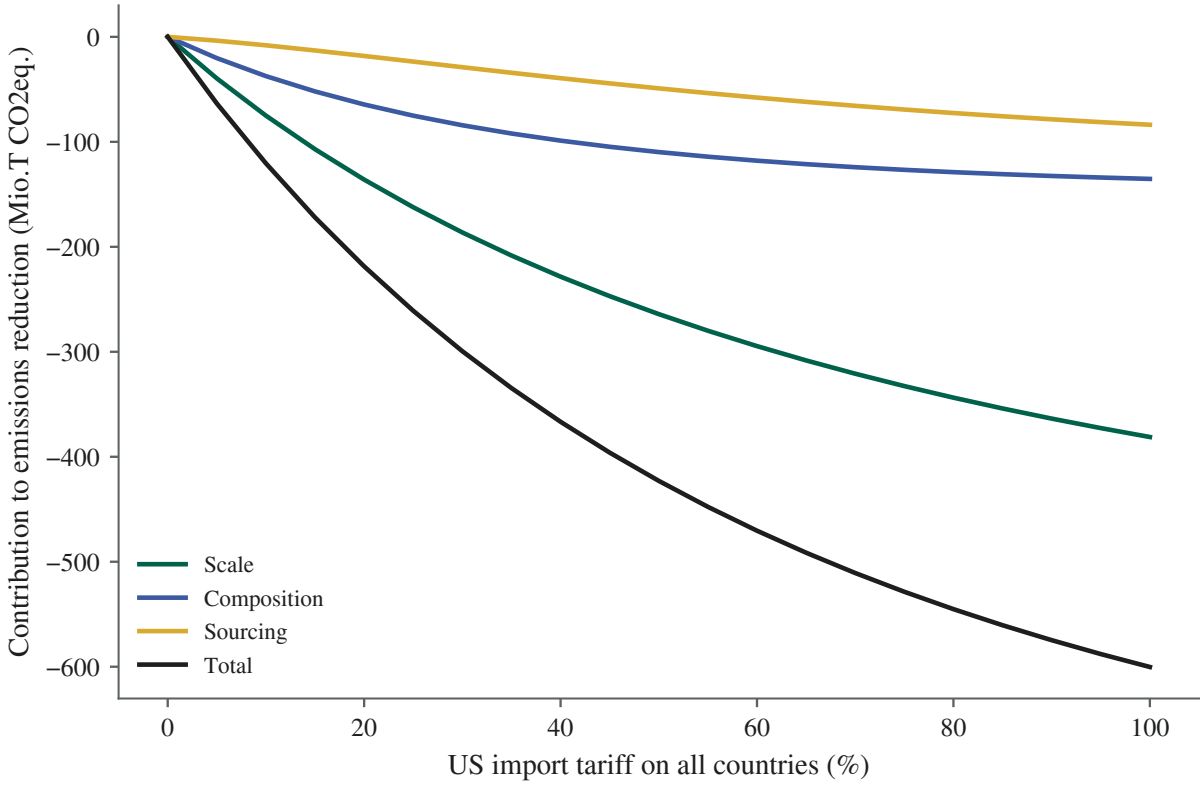
Figure 15: Global emission changes



Notes: Total change in global greenhouse gas emissions relative to the January 2025 baseline, in Mt CO_2eq., for each scenario. These totals correspond to the sums reported alongside the decomposition in Figure 5. Emissions intensities are held at baseline values throughout.

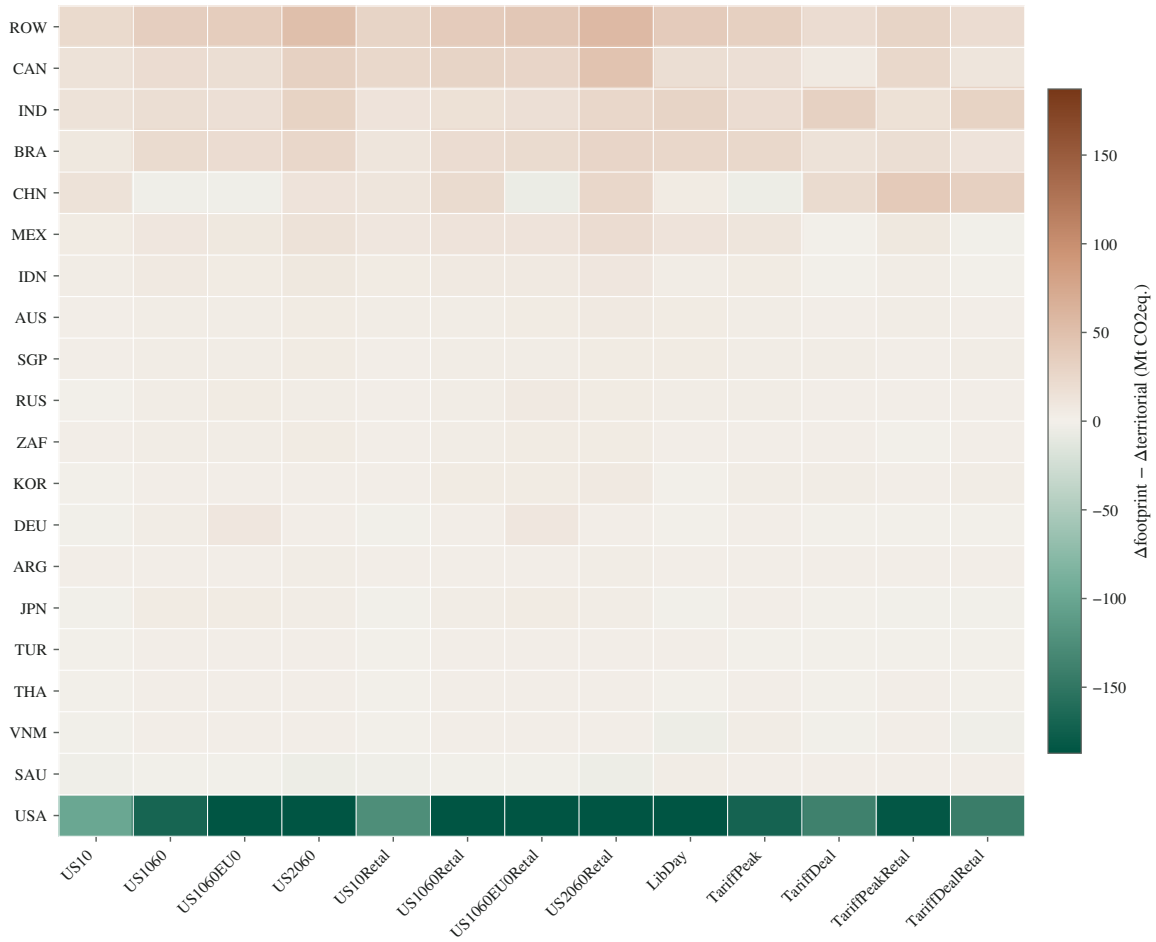
J Additional Graphs

Figure 16: Global emission changes - Varying U.S. tariff levels



Notes: Contribution of the scale, composition, and sourcing terms, together with their total, to the change in global emissions in Mt CO_2eq., as a uniform ad valorem U.S. tariff applied to all countries and all sectors rises from 0 to 100%. No partner retaliates.

Figure 17: Heatmap territorial vs consumption emissions



Notes: Net redistribution of embodied emissions, measured as the change in a country's carbon footprint less the change in its territorial emissions (Δfootprint − Δterritorial) in Mt CO_2eq., for each economy and each scenario. Brown cells indicate a positive gap, meaning the country absorbs embodied emissions displaced from elsewhere; teal cells indicate a negative gap. Rows are the twenty largest baseline territorial emitters plus the ICIO rest-of-world region, ordered as in Figure 8. As in that figure, the gap is a net magnitude and rises mechanically for large exporters when U.S. import demand contracts, so it should be read as a summary of redistribution rather than as direct evidence of dirtier sourcing.